

PRODUCT DATA		
PN: 8012420 / INT-034 – 100 L Seed Bioreactor (GPF-N, 500L Scale, Process 1)		
MPR Process Order No.:	General Supplies Process Order No.(s):	Recorded By/Date:

This record is an accurate reproduction of the Master.	Issued By:	Date Issued:
---	-------------------	---------------------

Completed Record Reviewed By:	Production:	Date:
	Quality Assurance:	Date:

1 Introduction

1.1 Purpose and Scope

1.1.1 This document describes the Intensified Fed Batch - 100 L Seed Bioreactor process in the Gaithersburg Pilot Facility-North (GPF-N).

1.2 General Instructions for Master Production Records

1.2.1 All personnel shall follow gowning requirements and personnel flows per SOP-0080154.

1.2.2 All equipment and materials shall flow through the GPF-N according to SOP-0080155.

1.2.3 A Batch ID is assigned by PCS.

2 Referenced Documents

2.1	SOP-0080273	Operation of Advanced MicroOsmometer
2.2	SOP-0080295	DLIMS Sample Submission for Manufacturing Sciences
2.3	SOP-0115565	Operation of the Beckman Coulter Vi-Cell BLU
2.4	SOP-0107252	Siemens BGA Operation
2.5	SOP-0080519	Gaithersburg Pilot Facility Upstream Viral Contamination Response Procedure
2.6	SOP-0107438	Biological Safety Cabinet Operation
2.7	SOP-0080812	C'eal-Flex TPE Ultra Sealer
2.8	SOP-0080522	Operation, Cleaning and Maintenance of the Vante TPE Tubing Sealer Model 4160
2.9	SOP-0080297	Benchtop pH and Conductivity Meter Operation
2.10	SOP-0080154	General Gowning Procedure
2.11	SOP-0107219	General Cleaning of Equipment, Supplies, and Materials for Entry into GPF-North
2.12	SOP-0080249	Operation, Cleaning, and Maintenance of Scales and Balances
2.13	SOP-0080257	Operation and Maintenance of the Sartorius BioWelder
2.14	SOP-0080224	Operation of Floor Scales
2.15	SOP-0107414	Operation of Masterflex Pumps
2.16	SOP-0107187	Operation of the Terumo SCD 11B Sterile Tubing Welder
2.17	SOP-0126420	Operation and Maintenance of the 100L ABEC CSR at the Gaithersburg Pilot Facility-North
2.18	SOP-0126879	Single-Use Tangential Flow Filter Operations for Integrated Dynamic Bioprocessing
2.19	SOP-0126942	Feed Cart Operations at GPF-North
2.20	SOP-0080155	General GPF-N Facility Information and Flow
2.21	SOP-0107056	cGMP Labeling Practices for Controlled Materials, Equipment, and Product
2.22	SOP-0081641	Manufacturing Sciences SAP Usage at Gaithersburg Pilot Facilities
2.23	FORM-0122774	ABEC Custom Single Run (CSR) – Parent Document SOP-0122730
2.24	FORM-0122776	Tangential Flow Filter (TFF) Parent Document SOP-0126879
2.25	MABR-0027687 PN: 8012419 / INT-034	Inoculum Expansion (GPFN, 500 L Scale, Process 1)
2.26	MABR-0027689 PN: 8012421 / INT-034	500 L Production Bioreactor (GPFN, 500 L Scale, Process 1)

3 Attachments

- 3.1 Attachment 1: Sample Assembly Attachment and Sampling**
- 3.2 Attachment 2: Permeate Bag Replacement**
- 3.3 Attachment 3: Additional Exhaust Filter Assembly**
- 3.4 Attachment 4: Antifoam Addition**
- 3.5 Attachment 5: Non-Routine Sampling Record**
- 3.6 Attachment 6: Additional pH Monitoring - pH samples' information**
- 3.7 Attachment 7: Retain Samples**
- 3.8 Attachment 8: Additional Sampling Pages (N-2)**
- 3.9 Attachment 9: Additional Sampling Pages (N-1)**
- 3.10 Attachment 10: Comments Section**

4 Bill of Materials

4.1 Equipment/ Components		
Item Number	Quantity	Description
BRX-4513	1	100 L Bioreactor
	As Needed	Tubing Welder
		Tubing Sealer
		Scale/Balance
		Peristaltic Pump
		Benchtop pH/Conductivity Meter
	1	Vi-CELL BLU Automated Cell Counter
	1	BGA (RAPIDpoint® 500)
	1	Osmometer
6005457	1	Single-Use Bioreactor Bag – 100 L (Custom)
6005503	As Needed	3/8 Pump Runner
6005484		AQS Male-Purefit
6005501		1/8" Pump Runner w AQS
6005484		AQS Male-Purefit
6005500		1/16 Pump Runner w AQS
4102603		Antifoam 1L
4102604		Antifoam 5L
6002363		Sample Assembly
6005501		1/8" Pump Runner w AQS
6005483		AQS Female-Purefit
6005485		1/8 Perfusion Manifold
4102461		300L Bioprocess Bag

4.1 Equipment/Components (continued)		
Item Number	Quantity	Description
6002348	As Needed	Exhaust Trap Line Bottle
6002392		Harvest Bag Transfer Assembly
6002347		Exhaust Filter Assembly
6001963		Pump Assembly
6002393		Pump runner assembly (1/8" x 1/4")
4101924		Flask with Attached C-Flex and Acro-50 (500 mL)
6002363		Cell Culture Sample Assembly Sterile (60 mL syringe)
6000391		Custom Assembly - Sampling Bags with 10 mL Syringe (250 mL)

4.2 Raw Materials	
Part Number	Description
8012419	INT-034 - Inoculum Expansion

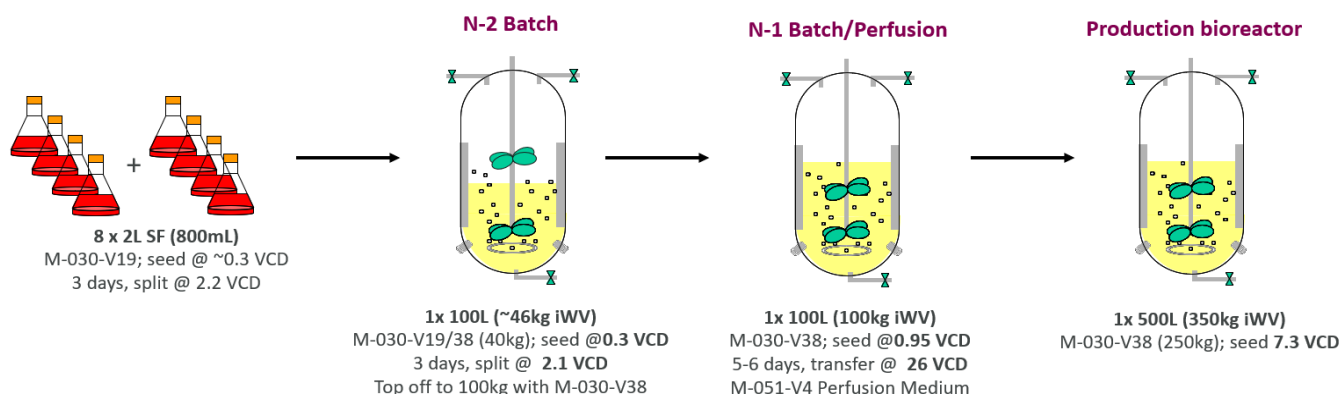
4.3 Solutions		
Part Number	Approximate Quantity	Description
8006887	40 kg	CCMB1 Medium (M-030-V19)
8010764	55 kg	CCMB1 Medium (M-030-V38)
8012342	500 kg	CCMB1 Perfusion Medium (M-051-V4)
4102603 4102604 OR 4102057	5 kg OR 2 kg	EX-CELL Antifoam OR Foam Away, Irradiated AOF, Sterile

5 Process Notes

5.1	The day of the inoculation is N-2 Seed Day 0. The day of medium dilution is N-1 Seed Day 0.
5.2	For the purpose of this record, 1 kg = 1 L, unless stated otherwise.
5.3	Record all decimals to the place indicated by the number of Xs, i.e., for "X.XX", record two decimal places and for "X.X", record to one decimal place.
5.4	Contact the area supervisor/designee if any of the operating conditions are outside of the specified range.
5.5	When proceeding with comments, the supervisor/designee is responsible for contacting the appropriate departments (i.e. Quality, CCFS-TT) to make an assessment. The process decisions need to be documented in Attachment 10: Comments Section .
5.6	Airflow rates and agitation speeds defined within this MPR are initial set points. These set points can be adjusted at the supervisor/designee's discretion. Record all changes in Attachment 10: Comments Section .
5.7	Airflow rates for both sparge and overlay and agitation speeds are defined as initial set points. These set points can be adjusted at the supervisor/designee's discretion, as long as the pO ₂ level is maintained within range. Record all changes on Attachment 10: Comments Section .
5.8	The initial pO ₂ alarm set points can be adjusted at the supervisor/designee's discretion, as long as the new alarm set points are within pO ₂ specified process limits. A supervisor/designee's signature is required and all changes need to be documented in Attachment 10: Comments Section .
5.9	Whenever the BGA is unavailable, measure pH levels using the Benchtop pH and Conductivity meter per SOP-0080297. Record the temperature compensated values denoted with (T), if applicable.
5.10	Minimize medium and nutrient feed exposure to light during storage or use.
5.11	Whenever sample bags/flasks are used for sampling, record the information in Attachment 1: Sample Assembly Attachment and Sampling .
5.12	Whenever a benchtop balance or floor scale is used, perform a weight check before use per SOP-0107155 or SOP-0080294.
5.13	Whenever a sealer is used, operate equipment per SOP-0080812 or SOP-0080522.
5.14	Additional samples for pH monitoring may be taken at supervisor/designee's discretion. Record the results in Attachment 6: Additional pH Monitoring - pH samples' information .
5.15	All Sections leading up to Media Charge and TFF Priming operation can be performed concurrently.

6 Process Flow Diagram

Seed (N-2) Phase 46 L Working Volume Seed at $\geq 0.3 \times 10^6$ viable cells/mL		
Medium Charge	40 kg of medium PN:8006887 CCMB1 Medium (M-030-V19)	
Medium Dilution (N-1) 100 L Working Volume Seed at $\geq 0.95 \times 10^6$ viable cells/mL		
Dilution Trigger	$\geq 2.1 \times 10^6$ viable cells/mL	
Dilution Medium	55 kg of medium PN: 8010764 CCMB1 Medium (M-030-V38)	
Perfusion Start (N-1)		
Perfusion Phase Trigger	$\geq 1.8 \times 10^6$ viable cells/mL	
	48 $\pm$ 6 hours after dilution	
Feed Cart Recipe Name	IFBN1 PermG	
Addition Strategy Flow Rates		
Solution	Flow Rates	
Antifoam	Addition as needed	
PN: 8012342 Perfusion Medium CCM-FR-2 (M-051-V4)	Hours Post Dilution: 48 – 72	0.3 VVD
	Hours Post Dilution: 72 – 96	0.70 VVD
	Hours Post Dilution: 96 – 120	1.50 VVD
	Hours Post Dilution: 120 - end	2.00 VVD
NOTE: Time range for perfusion rate changes have a $\pm$ 6 hour window on either end.		
Transfer to 500 L Production STR		
VCD Trigger	26 x 10 ⁶ viable cells/mL	



7 Bioreactor Set-Up

Step	Description	Performed By/Date	Witnessed By/Date						
7.1	Label BRX-4513 status tag with "INT-034 ", 8012420 , and batch number, initials and date.								
7.2	If maintenance has been performed, verify that the Start Up procedure has been performed per SOP-0126420.								
7.3	Verify that the pre-use cleaning of the CSR has been performed on FORM-0130911 per SOP-0126420.								
7.4	- Obtain the Disposable Container (DC) (6005457). - Record the Batch and expiry below.								
	<table border="1"> <thead> <tr> <th>Part Number</th><th>Batch Number</th><th>Exp. Date</th></tr> </thead> <tbody> <tr> <td></td><td></td><td></td></tr> </tbody> </table>	Part Number	Batch Number	Exp. Date					
Part Number	Batch Number	Exp. Date							
SAP Consumption Performed by/Date									
7.5	- Set up the DC per SOP-0126420. - Connect all exhaust, overlay and sparge filters.								
7.6	- Complete the Docking procedure of the DC per SOP-0126420. - Record the HMI Batch ID entered in the system below.								
	<table border="1"> <tr> <td>HMI Batch ID</td><td></td></tr> </table>	HMI Batch ID							
HMI Batch ID									
7.7	Ensure all major creases have been removed through air inflation.								
7.8	- Confirm the pressure hold test set points per SOP-0126420. - Perform an automated pressure hold test of the DC per SOP-0126420 and record required information in FORM-0130911. - If the P-Test on the DC after a third test, contact a supervisor and document the actions taken in the Attachment 10: Comments Section								
	<table border="1"> <tr> <td rowspan="5">DC Pressure Test Result</td><td>Test #1</td><td>Pass/Fail _____</td></tr> <tr> <td>Test #2</td><td>Pass/Fail _____ ☐ N/A</td></tr> <tr> <td>Test #3</td><td>Pass/Fail _____ ☐ N/A</td></tr> </table>	DC Pressure Test Result	Test #1	Pass/Fail _____	Test #2	Pass/Fail _____ ☐ N/A	Test #3	Pass/Fail _____ ☐ N/A	
DC Pressure Test Result	Test #1		Pass/Fail _____						
	Test #2		Pass/Fail _____ ☐ N/A						
	Test #3		Pass/Fail _____ ☐ N/A						

Step	Description	Performed By/Date	Witnessed By/Date																
7.9	Obtain and weld Cell Culture Sample Assembly (Y-Assembly) (6002363).																		
	<table border="1"> <thead> <tr> <th>Description</th><th>Part Number</th><th>Batch Number</th><th>Exp Date:</th><th>Quantity</th></tr> </thead> <tbody> <tr> <td>Cell Culture Sample Assembly (Y-Assembly)</td><td>6002363</td><td></td><td></td><td></td></tr> </tbody> </table>			Description	Part Number	Batch Number	Exp Date:	Quantity	Cell Culture Sample Assembly (Y-Assembly)	6002363									
	Description			Part Number	Batch Number	Exp Date:	Quantity												
	Cell Culture Sample Assembly (Y-Assembly)			6002363															
	Welder ID:																		
SAP Consumption Performed by/Date																			
7.10	- If applicable, obtain the following materials and record information below. - Weld the Exhaust Trap Line Bottle to the DC per SOP-0080257.																		
	<table border="1"> <thead> <tr> <th>Description</th><th>Part Number</th><th>Batch Number</th><th>Exp. Date</th></tr> </thead> <tbody> <tr> <td>Exhaust Trap Line Bottle</td><td>6002348</td><td></td><td></td></tr> <tr> <td>Harvest Transfer Assembly</td><td>6002392</td><td></td><td></td></tr> <tr> <td>3/8 to 1/4 Pump Assembly</td><td>6001963</td><td></td><td></td></tr> </tbody> </table>			Description	Part Number	Batch Number	Exp. Date	Exhaust Trap Line Bottle	6002348			Harvest Transfer Assembly	6002392			3/8 to 1/4 Pump Assembly	6001963		
	Description			Part Number	Batch Number	Exp. Date													
	Exhaust Trap Line Bottle			6002348															
	Harvest Transfer Assembly			6002392															
	3/8 to 1/4 Pump Assembly			6001963															
Welder ID:																			
7.11	- Per SOP-0126420, install and connect the probes and cables listed below into the probe belt of the ABEC DC: 2 Hamilton pH probes 2 Hamilton DO probes 1 Thermowell temperature probe - Record the Transmitter location and Serial number for each probe. NOTE: The temperature probe does not have a serial number or dedicated DC transmitter.																		
	<table border="1"> <thead> <tr> <th>Probe label</th><th>DC Transmitter location</th><th>Probe Serial No.</th></tr> </thead> <tbody> <tr> <td>pH Probe 1</td><td></td><td></td></tr> <tr> <td>pH Probe 2</td><td></td><td></td></tr> <tr> <td>DO Probe 1</td><td></td><td></td></tr> <tr> <td>DO Probe 2</td><td></td><td></td></tr> </tbody> </table>			Probe label	DC Transmitter location	Probe Serial No.	pH Probe 1			pH Probe 2			DO Probe 1			DO Probe 2			
	Probe label			DC Transmitter location	Probe Serial No.														
	pH Probe 1																		
	pH Probe 2																		
	DO Probe 1																		
	DO Probe 2																		

Step	Description		Performed By/Date	Witnessed By/Date
7.12	On the Loops Screen, set the following parameters per SOP-0126420.			
	Parameter	Set Point		
	Air Overlay	2.0 SLPM		
	Exhaust Filter Heater	70°C		
	Exhaust Deadband	0°C		
7.13	- Input the initial pH probe calibration values per SOP-0126420. - Record the Slope inputted into the HMI.			
	Probe	Slope (mV/pH)		
	1			
	2			

8 Process Recipe and Alarm Configuration

Step	Description	Performed By/Date	Witnessed By/Date																																								
This section can be completed at any time, prior to Section 11.																																											
8.1	For the following steps set up the Recipe Set points on the ABEC HMI through Recipe Manager.																																										
8.2	Set up the Recipe Tuning with the following parameters: <table><tr><td>Loop</td><td>P</td><td>I</td><td>D</td></tr><tr><td colspan="4">Temperature Tuning</td></tr><tr><td>Vessel Temp</td><td>5.0</td><td>0.010</td><td>0.100</td></tr><tr><td colspan="4">pH Tuning</td></tr><tr><td>Primary CO2 Sparge</td><td>0.08</td><td>0.030</td><td>0.000</td></tr><tr><td>Base Pump</td><td>0.000</td><td>0.000</td><td>0.000</td></tr><tr><td colspan="4">Level Tuning</td></tr><tr><td></td><td>4.000</td><td>0.900</td><td>0.000</td></tr><tr><td colspan="4">DO Profile Control Tuning</td></tr><tr><td>DO Control</td><td>0.010</td><td>0.010</td><td>0.000</td></tr></table>	Loop	P	I	D	Temperature Tuning				Vessel Temp	5.0	0.010	0.100	pH Tuning				Primary CO2 Sparge	0.08	0.030	0.000	Base Pump	0.000	0.000	0.000	Level Tuning					4.000	0.900	0.000	DO Profile Control Tuning				DO Control	0.010	0.010	0.000		
Loop	P	I	D																																								
Temperature Tuning																																											
Vessel Temp	5.0	0.010	0.100																																								
pH Tuning																																											
Primary CO2 Sparge	0.08	0.030	0.000																																								
Base Pump	0.000	0.000	0.000																																								
Level Tuning																																											
	4.000	0.900	0.000																																								
DO Profile Control Tuning																																											
DO Control	0.010	0.010	0.000																																								
8.3	Set the DO Profile Output Tuning: Select the checkboxes for Agitation and Primary O2 Sparge. Input the DO Cascade Strategy set points below: <table><tr><td colspan="3">DO Profile Setpoints</td></tr><tr><td>DO Output %</td><td>Agitation (rpm)</td><td>Primary O2 Sparge (slpm)</td></tr><tr><td>0</td><td>75</td><td>0.0</td></tr><tr><td>20</td><td>80</td><td>2.0</td></tr><tr><td>40</td><td>90</td><td>4.0</td></tr><tr><td>60</td><td>105</td><td>6.0</td></tr><tr><td>80</td><td>120</td><td>8.0</td></tr><tr><td>100</td><td>140</td><td>10.0</td></tr></table>	DO Profile Setpoints			DO Output %	Agitation (rpm)	Primary O2 Sparge (slpm)	0	75	0.0	20	80	2.0	40	90	4.0	60	105	6.0	80	120	8.0	100	140	10.0																		
DO Profile Setpoints																																											
DO Output %	Agitation (rpm)	Primary O2 Sparge (slpm)																																									
0	75	0.0																																									
20	80	2.0																																									
40	90	4.0																																									
60	105	6.0																																									
80	120	8.0																																									
100	140	10.0																																									

Step	Description	Performed By/Date	Witnessed By/Date																		
8.4	On the Recipe Tuning page, select DO Auto Tuning: - Under DO Setpoints “Cascade Strategy”, set option 1 to “Primary O2 Sparge”. Ensure options 2 - 4 are set to “None”. - Set the DO Tuning parameters per the table below: <table><tr><th>Loop</th><th>P</th><th>I</th><th>D</th></tr><tr><td colspan="4">DO Tuning</td></tr><tr><td>Primary O2 Sparge</td><td>0.001</td><td>0.001</td><td>0.000</td></tr></table> - Set the Primary O2 Sparge Cascade Ranges per the table below: <table><tr><th>Loop</th><th>Min (slpm)</th><th>Max (slpm)</th></tr><tr><td>Primary O2 Sparge</td><td>0</td><td>20</td></tr></table>	Loop	P	I	D	DO Tuning				Primary O2 Sparge	0.001	0.001	0.000	Loop	Min (slpm)	Max (slpm)	Primary O2 Sparge	0	20		
Loop	P	I	D																		
DO Tuning																					
Primary O2 Sparge	0.001	0.001	0.000																		
Loop	Min (slpm)	Max (slpm)																			
Primary O2 Sparge	0	20																			

Step	Description	Performed By/Date	Witnessed By/Date																										
8.5	Under Recipe Set Points, input the Recipe Set Points for the Charge step with the following parameters: <table><tr><th colspan="2">Charge Recipe</th></tr><tr><td>Vessel Temp SP</td><td>36.5 °C</td></tr><tr><td>Primary Air Sparge SP</td><td>3.1 slpm</td></tr><tr><td>Air Overlay SP</td><td>5.0 slpm</td></tr><tr><td>Primary CO2 Sparge SP</td><td>0.2 slpm</td></tr><tr><td>Agitation On Weight CP</td><td>30 kg</td></tr><tr><td>Agitation On SP</td><td>75 rpm</td></tr><tr><td>Agitation Ramp UP SP</td><td>75 rpm</td></tr><tr><td>Pre-Fill Weight CP</td><td>35 kg</td></tr><tr><td>Main Fill Weight CP</td><td>40 kg</td></tr></table>	Charge Recipe		Vessel Temp SP	36.5 °C	Primary Air Sparge SP	3.1 slpm	Air Overlay SP	5.0 slpm	Primary CO2 Sparge SP	0.2 slpm	Agitation On Weight CP	30 kg	Agitation On SP	75 rpm	Agitation Ramp UP SP	75 rpm	Pre-Fill Weight CP	35 kg	Main Fill Weight CP	40 kg								
Charge Recipe																													
Vessel Temp SP	36.5 °C																												
Primary Air Sparge SP	3.1 slpm																												
Air Overlay SP	5.0 slpm																												
Primary CO2 Sparge SP	0.2 slpm																												
Agitation On Weight CP	30 kg																												
Agitation On SP	75 rpm																												
Agitation Ramp UP SP	75 rpm																												
Pre-Fill Weight CP	35 kg																												
Main Fill Weight CP	40 kg																												
8.6	Under Recipe Set Points, input the Recipe Set Points for the Growth Stabilize phase with the following parameters: <table><tr><th colspan="2">Growth Stabilize</th></tr><tr><td>Vessel Temp SP</td><td>36.5 °C</td></tr><tr><td>pH SP</td><td>7.35</td></tr><tr><td>pH deadband</td><td>0.10</td></tr><tr><td>DO SP</td><td>100%</td></tr><tr><td>Agitation SP</td><td>75 rpm</td></tr><tr><td>Air Overlay SP</td><td>5.0 slpm</td></tr><tr><td>Primary Air Sparge SP</td><td>0.5 slpm</td></tr><tr><td>Secondary Air Sparge SP</td><td>0</td></tr><tr><td>Primary O2 Sparge SP</td><td>0</td></tr><tr><td>Secondary O2 Sparge SP</td><td>0</td></tr><tr><td>pH Strategy</td><td>Auto CO2</td></tr><tr><td>DO Strategy</td><td>OFF</td></tr></table>	Growth Stabilize		Vessel Temp SP	36.5 °C	pH SP	7.35	pH deadband	0.10	DO SP	100%	Agitation SP	75 rpm	Air Overlay SP	5.0 slpm	Primary Air Sparge SP	0.5 slpm	Secondary Air Sparge SP	0	Primary O2 Sparge SP	0	Secondary O2 Sparge SP	0	pH Strategy	Auto CO2	DO Strategy	OFF		
Growth Stabilize																													
Vessel Temp SP	36.5 °C																												
pH SP	7.35																												
pH deadband	0.10																												
DO SP	100%																												
Agitation SP	75 rpm																												
Air Overlay SP	5.0 slpm																												
Primary Air Sparge SP	0.5 slpm																												
Secondary Air Sparge SP	0																												
Primary O2 Sparge SP	0																												
Secondary O2 Sparge SP	0																												
pH Strategy	Auto CO2																												
DO Strategy	OFF																												

Step	Description	Performed By/Date	Witnessed By/Date
8.7	Under Recipe Set Points, input the recipe set points for the Growth Inoculate/Grow 1 (N-2) phase:		
	Growth Inoculate/Grow 1 (N-2)		
	Vessel Temp SP		
	pH SP		
	pH deadband		
	DO SP		
	Agitation SP		
	Air Overlay SP		
	Primary Air Sparge SP		
	Secondary Air Sparge SP		
	Primary O2 Sparge SP		
	Secondary O2 Sparge SP		
	pH Strategy		
	DO Strategy		

Step	Description	Performed By/Date	Witnessed By/Date
8.8	Under Recipe Set Points, input the recipe set points for the Growth Grow 2 (N-1) :		
	Growth Grow 2 (N-1)		
	Vessel Temp SP	36.5 °C	
	pH SP	7.10	
	pH deadband	0.20	
	DO SP	50%	
	Agitation SP	75 rpm	
	Air Overlay SP	5.0 slpm	
	Primary Air Sparge SP	0.5 slpm	
	Secondary Air Sparge SP	0	
	Primary O2 Sparge SP	0	
	Secondary O2 Sparge SP	0	
	pH Strategy	Auto CO2	
	DO Strategy	Profile	

Step	Description	Performed By/Date	Witnessed By/Date																																																						
8.9	For the Following steps set up the alarms on the ABEC HMI.																																																								
8.10	Under the Recipe Alarms, input the follow ABEC System Alarms: <table><tr><th colspan="6">Bioreactor System Alarms</th></tr><tr><th></th><th>Low Interlock</th><th>Low</th><th>High</th><th>High Interlock</th><th>Variance</th></tr><tr><td>Vessel Temp (°C)</td><td></td><td></td><td></td><td>38.5</td><td></td></tr><tr><td>Vessel Pressure (psi)</td><td></td><td></td><td>0.4</td><td>0.45</td><td></td></tr><tr><td>Vessel Weight (kg)</td><td>0</td><td></td><td></td><td>130</td><td></td></tr><tr><td>Agitation (rpm)</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>pH</td><td>6.2</td><td></td><td></td><td>7.7</td><td>0.2</td></tr><tr><td>DO (%)</td><td></td><td></td><td></td><td></td><td>20</td></tr><tr><td>Auxiliary DO (%)</td><td></td><td>0</td><td>200</td><td></td><td></td></tr></table>	Bioreactor System Alarms							Low Interlock	Low	High	High Interlock	Variance	Vessel Temp (°C)				38.5		Vessel Pressure (psi)			0.4	0.45		Vessel Weight (kg)	0			130		Agitation (rpm)						pH	6.2			7.7	0.2	DO (%)					20	Auxiliary DO (%)		0	200				
Bioreactor System Alarms																																																									
	Low Interlock	Low	High	High Interlock	Variance																																																				
Vessel Temp (°C)				38.5																																																					
Vessel Pressure (psi)			0.4	0.45																																																					
Vessel Weight (kg)	0			130																																																					
Agitation (rpm)																																																									
pH	6.2			7.7	0.2																																																				
DO (%)					20																																																				
Auxiliary DO (%)		0	200																																																						
8.11	Under the Recipe Alarms, input the follow Setup/Charge/Stabilize Alarms: <table><tr><th colspan="4">Setup/Charge/Stabilize Alarm</th></tr><tr><th></th><th>Low</th><th>Deviation</th><th>High</th></tr><tr><td>Vessel Temp (°C)</td><td>35.0</td><td>1.0</td><td>38.0</td></tr><tr><td>Vessel Weight (kg)</td><td>30</td><td>5</td><td>50</td></tr><tr><td>Agitation (rpm)</td><td>65</td><td></td><td>85</td></tr><tr><td>pH</td><td>6.85</td><td>0.11</td><td>7.50</td></tr><tr><td>DO (%)</td><td>10</td><td>10</td><td>110</td></tr></table>	Setup/Charge/Stabilize Alarm					Low	Deviation	High	Vessel Temp (°C)	35.0	1.0	38.0	Vessel Weight (kg)	30	5	50	Agitation (rpm)	65		85	pH	6.85	0.11	7.50	DO (%)	10	10	110																												
Setup/Charge/Stabilize Alarm																																																									
	Low	Deviation	High																																																						
Vessel Temp (°C)	35.0	1.0	38.0																																																						
Vessel Weight (kg)	30	5	50																																																						
Agitation (rpm)	65		85																																																						
pH	6.85	0.11	7.50																																																						
DO (%)	10	10	110																																																						

Step	Description	Performed By/Date	Witnessed By/Date
8.12	Under the Recipe Alarms, input the Inoculate/Grow 1 (N-2) Alarms:		
	Inoculate/Grow 1 (N-2) Alarms		
	Low	Deviation	High
	Vessel Temp (°C)	35.0	1.0
	Vessel Weight (kg)	35	5
	Agitation (rpm)	60	150
	pH	6.85	0.11
8.13	Under the Recipe Alarms, input the Grow 2 (N-1) Alarms:		
	Grow 2 (N-1)		
	Low	Deviation	High
	Vessel Temp (°C)	35.0	1.0
	Vessel Weight (kg)	85	10
	Agitation (rpm)	60	150
	pH	6.70	0.22
8.14	Under the Recipe Alarms, input the Levitronix 1 Alarms:		
	Levitronix 1 Alarms		
		Low	High
	Pump Speed (RPM)	600	3000
	Flow rate (L/hr)	2.4	6.4
	Temp (°C)		
	TMP (psi)	0	0.5
	P1 (psi)	0	3
	P2 (psi)	0	3
	P3 (psi)	0	3

9 TFF Set up

Step	Description	Performed By/Date	Witnessed By/Date										
9.1	Obtain 1 TFF Filter assembly:												
	<table border="1"> <thead> <tr> <th>Description</th><th>Part Number</th><th>Serial No.</th><th>Batch Number</th><th>Expiry Date</th></tr> </thead> <tbody> <tr> <td>TFF 1</td><td>6005470</td><td></td><td></td><td></td></tr> </tbody> </table>			Description	Part Number	Serial No.	Batch Number	Expiry Date	TFF 1	6005470			
	Description			Part Number	Serial No.	Batch Number	Expiry Date						
	TFF 1	6005470											
SAP Consumption Performed by/Date													
9.2	Verify per FORM-0122776 that TFF Filter Holder pre-use cleaning has been performed.												
9.3	- Perform a P-Test on the filter in Step 9.1 per SOP-0126879. - If the P-Test on the filter fails after a second test, contact a supervisor and document the actions taken in the Attachment 10: Comments Section. - Utilize the following P-Test criteria.												
	<table border="1"> <thead> <tr> <th>Pressurize the TFF</th><th>Stabilization Time</th><th>Minimum P-Test hold time</th><th>Acceptable TFF pressure loss</th></tr> </thead> <tbody> <tr> <td>5 psi</td><td>10 min</td><td>30 min</td><td>≤ 0.10 psi</td></tr> </tbody> </table>			Pressurize the TFF	Stabilization Time	Minimum P-Test hold time	Acceptable TFF pressure loss	5 psi	10 min	30 min	≤ 0.10 psi		
	Pressurize the TFF			Stabilization Time	Minimum P-Test hold time	Acceptable TFF pressure loss							
5 psi	10 min	30 min	≤ 0.10 psi										
9.4	Record the P-Test result of the TFF filter below and record required information on FORM-0122776.												
	<table border="1"> <thead> <tr> <th>TFF 1 Pressure Test Result</th><th>Test #1</th><th>Pass/Fail _____</th></tr> </thead> <tbody> <tr> <td></td><td>Test #2</td><td>Pass/Fail _____ ☐ N/A</td></tr> </tbody> </table>			TFF 1 Pressure Test Result	Test #1	Pass/Fail _____		Test #2	Pass/Fail _____ ☐ N/A				
	TFF 1 Pressure Test Result			Test #1	Pass/Fail _____								
	Test #2	Pass/Fail _____ ☐ N/A											
9.5	If needed, obtain the following material for TFF Set Up:												
	<table border="1"> <thead> <tr> <th>Description</th><th>Part Number</th><th>Batch Number</th><th>Expiry Date</th><th>Quantity</th></tr> </thead> <tbody> <tr> <td>1' G-to-G Jumper</td><td>6005498</td><td></td><td></td><td></td></tr> </tbody> </table>			Description	Part Number	Batch Number	Expiry Date	Quantity	1' G-to-G Jumper	6005498			
	Description			Part Number	Batch Number	Expiry Date	Quantity						
	1' G-to-G Jumper	6005498											
SAP Consumption Performed by/Date													

Step	Description	Performed By/Date	Witnessed By/Date									
9.6	- Set up the TFF per SOP-0126879. - Record the Levitronix pump ID attached to the filter.											
	<table border="1"> <tr> <td>TFF Filter</td><td>Levitronix Pump</td></tr> </table>			TFF Filter	Levitronix Pump							
	TFF Filter			Levitronix Pump								
<table border="1"> <tr> <td>TFF 1</td><td></td></tr> </table>	TFF 1											
TFF 1												
9.7	- Perform a system P-Test of the TFF System loop per SOP-0126879. - If the P-Test for the system fails after a second test, contact a supervisor and document the actions taken in the Attachment 10: Comments Section. - Utilize the following parameters for the system P-Test:											
	<table border="1"> <tr> <td>Pressurize the TFF</td><td>Stabilization Time</td><td>Minimum Stabilization hold time</td><td>Acceptable TFF pressure loss</td></tr> <tr> <td>5 psi</td><td>10 min</td><td>30 min</td><td>≤ 0.10 psi</td></tr> </table>			Pressurize the TFF	Stabilization Time	Minimum Stabilization hold time	Acceptable TFF pressure loss	5 psi	10 min	30 min	≤ 0.10 psi	
	Pressurize the TFF			Stabilization Time	Minimum Stabilization hold time	Acceptable TFF pressure loss						
5 psi	10 min	30 min	≤ 0.10 psi									
9.8	Record the P-Test result of the TFF System below:											
	<table border="1"> <tr> <td rowspan="2">TFF System P-Test Result</td><td>Test #1</td><td>Pass/Fail _____</td></tr> <tr> <td>Test #2</td><td>Pass/Fail _____ ☐ N/A</td></tr> </table>			TFF System P-Test Result	Test #1	Pass/Fail _____	Test #2	Pass/Fail _____ ☐ N/A				
	TFF System P-Test Result				Test #1	Pass/Fail _____						
Test #2		Pass/Fail _____ ☐ N/A										

10 Connections to Bioreactor

Step	Description	Performed By/Date	Witnessed By/Date																																			
10.1	Obtain the material needed to set up the permeate line and waste bag																																					
	<table border="1"> <thead> <tr> <th>Description</th><th>Part Number</th><th>Batch Number</th><th>Expiry Date</th><th>Quantity</th></tr> </thead> <tbody> <tr> <td>1/8" Pump Runner with AQS</td><td>6005501</td><td></td><td></td><td></td></tr> <tr> <td rowspan="2">S-to-S Jumper</td><td>☐ 6005492</td><td></td><td></td><td></td></tr> <tr> <td>☐ 6005491</td><td></td><td></td><td></td></tr> <tr> <td rowspan="2">Bioprocess Bag</td><td>☐ 4102462</td><td></td><td></td><td></td></tr> <tr> <td>☐ 4102461</td><td></td><td></td><td></td></tr> <tr> <td>Purefit Female to AQS Adapter</td><td>6005483*</td><td></td><td></td><td></td></tr> </tbody> </table>			Description	Part Number	Batch Number	Expiry Date	Quantity	1/8" Pump Runner with AQS	6005501				S-to-S Jumper	☐ 6005492				☐ 6005491				Bioprocess Bag	☐ 4102462				☐ 4102461				Purefit Female to AQS Adapter	6005483*					
	Description			Part Number	Batch Number	Expiry Date	Quantity																															
	1/8" Pump Runner with AQS			6005501																																		
	S-to-S Jumper			☐ 6005492																																		
				☐ 6005491																																		
	Bioprocess Bag			☐ 4102462																																		
		☐ 4102461																																				
	Purefit Female to AQS Adapter	6005483*																																				
*If PN 6005483 is not available, weld the permeate waste bag in Step 10.2 .																																						
SAP Consumption Performed by/Date																																						
10.2	Set up the permeate line and waste bag per SOP-0126420.																																					
	Welder ID, if applicable																																					
10.3	Label the permeate bag with "INT-034 Permeate Waste", 8012420 , Batch Number, Initials/Date.																																					
10.4	Obtain the net weight of the permeate vessel after attaching to the DC.																																					
	Scale ID																																					
	Calibration Due Date																																					
	Tare Weight of Permeate Vessel			kg																																		

Step	Description	Performed By/Date	Witnessed By/Date	
Steps 10.5 - 10.7 can be done at any time prior to Section 15 .				
10.5	- Obtain all the material needed to set up the connection to the Antifoam bag. - Record the batch number, expiry and quantity used.			
	Description	Part Number	Batch No.	
	Antifoam	☐ 4102603		
		☐ 4102604		
		☐ 4102057		
	1/8" Pump Runner	6002393		
SAP Consumption Performed by/Date				
10.6	Weld the antifoam bag to the DC per SOP-0107187 and record welder information below:			
	Welder ID			
10.7	- Tare the balance. - Place the antifoam bag on the balance. - Record the weight of the antifoam bag before priming. - Prime the antifoam line per SOP-0080233. - Record the weight of the antifoam bag after priming.			
	Scale / Balance ID:			
	Calibration Due Date			
	Weight of Antifoam Bag (X)			
	Weight before priming			
	Weight after priming			
	Configuration for Tare weight	☐ weigh with tubing ☐ weigh inside container		

Step	Description	Performed By/Date	Witnessed By/Date												
All Steps past this point can be done at any time prior to Section 18 .															
10.8	- Obtain the materials needed to connect the DC to the Perfusion Medium line. - Record the batch number, expiry and quantity used.														
	<table border="1"> <thead> <tr> <th>Description</th><th>Part Number</th><th>Batch No.</th><th>Exp. Date</th><th>Quantity.</th></tr> </thead> <tbody> <tr> <td>1/8" Pump Runner with AQS</td><td>6005501</td><td></td><td></td><td></td></tr> <tr> <td>Male Purefit to AQS</td><td>6005484*</td><td></td><td></td><td></td></tr> </tbody> </table>			Description	Part Number	Batch No.	Exp. Date	Quantity.	1/8" Pump Runner with AQS	6005501				Male Purefit to AQS	6005484*
Description	Part Number	Batch No.	Exp. Date	Quantity.											
1/8" Pump Runner with AQS	6005501														
Male Purefit to AQS	6005484*														
*If PN 6005484 is not available, weld the Perfusion Medium in Step 10.10 .															
SAP Consumption Performed by/Date															
10.9	- Obtain CCM-FR-2 Perfusion Medium (M-051-V4) (PN: 8012342). - Record the Perfusion Medium information below.														
	Part Number:														
	Batch Number:														
	Expiration Date:														
	Scale ID:														
	Calibration Due Date:														
	Gross Weight of CCM-FR-2 Perfusion Medium (M-051-V4) (PN: 8012342) before use		kg												
10.10	Set up and connect the Perfusion Medium bag to the ABEC port per SOP-0126420.														
	Welder ID, if applicable:														

Step	Description	Performed By/Date	Witnessed By/Date	
10.11	<u>Perfusion Medium Sample:</u> - Obtain a sampling assembly (6000391). - Record the assembly and welder information. - Label this assembly with “INT-034”, 8012420 and batch number, bioreactor ID, “100 L Charge Medium”, initials, and date. - Weld the assembly to the Medium bag per SOP-0107187 or SOP-0107297.			
	Welder ID			
	Part No.:			
	Batch No.:			
	Exp. Date:			
	SAP Consumption Performed By/Date			
	10.12	- Dispense approximately 50 mL to be stored as investigational sample. - Seal off assembly for removal and record sealer information. - Record the sample collection in Attachment 7: Retain Samples.		
Sealer ID				

11 Media Charge and TFF Priming

Step	Description	Performed By/Date	Witnessed By/Date								
11.1	Ensure Sections 7 - 9 and Steps 10.1 - 10.4 are completed prior to starting Section 11 .										
11.2	Verify the process trends for BRX-4513 and TFF-4501 are activated.										
11.3	Obtain the following material and record the information below.										
	<table border="1"> <thead> <tr> <th>Description</th><th>Part No.</th><th>Batch No.</th><th>Expiry Date</th></tr> </thead> <tbody> <tr> <td>Male Purefit to AQS</td><td>6005484*</td><td></td><td></td></tr> </tbody> </table>	Description	Part No.	Batch No.	Expiry Date	Male Purefit to AQS	6005484*				
Description	Part No.	Batch No.	Expiry Date								
Male Purefit to AQS	6005484*										
	*If PN 6005484 is not available, weld the CCMB1 Medium in Step 11.4 .										
	SAP Consumption Performed By/Date										
11.4	- Obtain CCMB1 Medium (M-030-V19) (PN: 8006887). - Connect medium to the DC charge line per SOP-0162420. <table border="1"> <tbody> <tr> <td>Part No.:</td><td></td></tr> <tr> <td>Batch No.:</td><td></td></tr> <tr> <td>Exp. Date:</td><td></td></tr> <tr> <td>Welder ID, if applicable</td><td></td></tr> </tbody> </table>	Part No.:		Batch No.:		Exp. Date:		Welder ID, if applicable			
Part No.:											
Batch No.:											
Exp. Date:											
Welder ID, if applicable											
11.5	<u>Medium Sample:</u> - Obtain a sampling assembly (6000391). - Record the assembly and welder information. - Label this assembly with "INT-034", 8012420 and batch number, bioreactor ID, "100 L Charge Medium", initials, and date. - Weld the assembly to the Medium bag per SOP-0107187 or SOP-0107297. <table border="1"> <tbody> <tr> <td>Welder ID</td><td></td></tr> <tr> <td>Part No.:</td><td></td></tr> <tr> <td>Batch No.:</td><td></td></tr> <tr> <td>Exp. Date:</td><td></td></tr> </tbody> </table>	Welder ID		Part No.:		Batch No.:		Exp. Date:			
Welder ID											
Part No.:											
Batch No.:											
Exp. Date:											
	SAP Consumption Performed By/Date										

Step	Description				Performed By/Date	Witnessed By/Date
11.6	- Dispense approximately 50 mL to be stored as investigational sample. - Seal off assembly for removal and record sealer information. - Record the sample collection in Attachment 7: Retain Samples.					
	Sealer ID					
11.7	- Advance to the Charge phase on the ABEC HMI per SOP-0126420. - Verify the parameters align with Step 8.5.					
11.8	- Prime the DC medium charge line. - Tare the Bioreactor Weight. - Record the weight of the bioreactor after completing the tare.					
	Bioreactor Tare Weight		kg			
11.9	Transfer 40 ± 2 kg of medium (PN: 8006887) into the bioreactor per SOP-0126420.					
	Transfer Start time:	Transfer End time:	Amount Added:	Post Addition Exp. Date:		
			_____ (38 – 42 kg)	_____ (5 days after medium charge)		
SAP Consumption Performed By/Date						

Step	Description			Performed By/Date	Witnessed By/Date
11.10	NOTE: Do not proceed to the Growth Stabilize phase in the ABEC HMI. Prior to starting TFF priming, record the Bioreactor weight.				
	Bioreactor weight		kg		
11.11	Ensure the system stays in Manual Addition mode for the TFF Priming procedures.				
11.12	Ensure the TFF 1 permeate line is clamped and the clamp-on flow meter is installed per SOP-0126879.				
11.13	- Record the bioreactor weight pre-priming below. - Prime the TFF1 retentate line per SOP-0126879. - After fully priming the filter, record the post-prime BRX weight (excluding the permeate line).				
	TFF Description	BRX Weight Pre-Priming Weight	BRX Weight Post Prime		
	TFF 1	kg	kg		
11.14	Calculate the TFF 1 Hold up volume.				
	BRX Weight Pre-Priming (Step 11.13)	-	BRX Weight Post Prime (Step 11.13)	=	TFF 1 Hold Up Volume
	kg		kg		kg

Step	Description	Performed By/Date	Witnessed By/Date						
11.15	After Levitronix is turned off, per SOP-0126879: - Clamp the DC supply line to the TFF. - Clamp the DC return line.								
11.16	Zero the TFF 1 retentate clamp-on flow meter per SOP-0126879.								
11.17	Zero the TFF 1 pressure sensors per SOP-0126879.								
11.18	- Open the DC return line. - Open the DC supply line to the TFF. - Record the TFF Selection corresponding to the TFF set up in Step 9.6 in the table below - Turn on the TFF 1 Recirculation set points in the Levitronix per SOP-0126420. <table><tr><th>TFF Selection</th><th>Flow Rate</th><th>Control Selection</th></tr><tr><td>_____</td><td>4.4 lpm</td><td>Flow</td></tr></table>	TFF Selection	Flow Rate	Control Selection	_____	4.4 lpm	Flow		
TFF Selection	Flow Rate	Control Selection							
_____	4.4 lpm	Flow							
11.19	On the ABEC Loops Screen, ensure the temperature is set per the table below. <table><tr><th>Parameter</th><th>Set point</th></tr><tr><td>Vessel Temp SP</td><td>36.5 °C</td></tr></table>	Parameter	Set point	Vessel Temp SP	36.5 °C				
Parameter	Set point								
Vessel Temp SP	36.5 °C								

Step	Description	Performed By/Date	Witnessed By/Date														
11.20	Calculate the Target Bioreactor Weight needed for The TCCV target.																
	<table border="1"> <tr> <td>Target Media Charge weight prior to Inoculation</td><td>-</td><td>TFF 1 Hold up Volume (Step 11.14)</td><td>=</td><td>Target Bioreactor Weight</td></tr> <tr> <td>40 kg</td><td></td><td>kg</td><td></td><td>kg</td></tr> </table>			Target Media Charge weight prior to Inoculation	-	TFF 1 Hold up Volume (Step 11.14)	=	Target Bioreactor Weight	40 kg		kg		kg				
	Target Media Charge weight prior to Inoculation			-	TFF 1 Hold up Volume (Step 11.14)	=	Target Bioreactor Weight										
40 kg		kg		kg													
11.21	- Record the current bioreactor weight on the HMI. - Calculate the difference between the target bioreactor weight and the current bioreactor weight.																
	<table border="1"> <tr> <td>Current Bioreactor Weight (ABEC HMI)</td><td>-</td><td>Target Bioreactor Weight (Step 11.20)</td><td>=</td><td>Bioreactor weight difference</td></tr> <tr> <td>kg</td><td></td><td>kg</td><td></td><td>kg</td></tr> </table>			Current Bioreactor Weight (ABEC HMI)	-	Target Bioreactor Weight (Step 11.20)	=	Bioreactor weight difference	kg		kg		kg				
	Current Bioreactor Weight (ABEC HMI)			-	Target Bioreactor Weight (Step 11.20)	=	Bioreactor weight difference										
kg		kg		kg													
11.22	Check and complete the following activity based on the bioreactor weight difference.																
	<table border="1"> <tr> <th>Bioreactor weight difference condition</th><th>Action To take</th><th>Check the action completed</th></tr> <tr> <td>Step 11.21 > 0</td><td>Flush medium through TFF 1 permeate until the amount calculated in Step 11.20 is reached on the ABEC HMI.</td><td>☐</td></tr> <tr> <td>Step 11.21 < 0</td><td>Transfer the amount of CCMB1 Medium (M-030-V19) (PN: 8006887) to reach the calculated target bioreactor weight in Step 11.20.</td><td>☐</td></tr> <tr> <td>Step 11.21 = 0</td><td>Continue to the next step.</td><td>☐</td></tr> </table>			Bioreactor weight difference condition	Action To take	Check the action completed	Step 11.21 > 0	Flush medium through TFF 1 permeate until the amount calculated in Step 11.20 is reached on the ABEC HMI.	☐	Step 11.21 < 0	Transfer the amount of CCMB1 Medium (M-030-V19) (PN: 8006887) to reach the calculated target bioreactor weight in Step 11.20 .	☐	Step 11.21 = 0	Continue to the next step.	☐		
	Bioreactor weight difference condition			Action To take	Check the action completed												
	Step 11.21 > 0			Flush medium through TFF 1 permeate until the amount calculated in Step 11.20 is reached on the ABEC HMI.	☐												
Step 11.21 < 0	Transfer the amount of CCMB1 Medium (M-030-V19) (PN: 8006887) to reach the calculated target bioreactor weight in Step 11.20 .	☐															
Step 11.21 = 0	Continue to the next step.	☐															

Step	Description	Performed By/Date	Witnessed By/Date																		
11.23	Ensure that the temperature reading is within the process limits of 35.0 – 38.0 °C and that the pO₂ readings on the HMI are stable before proceeding to Step 11.24. NOTE: If readings are not stable after 4 hours, contact supervisor/designee and record provided instructions in Attachment 10: Comments Section. Supervisor/designee contacted: _____																				
11.24	Once the readings are stable, standardize the DO probe to 100% per SOP-0126420. <table border="1"> <tr> <td>Time 100% Standardization performed (AIT-01):</td><td></td></tr> <tr> <td>Time 100% Standardization performed (AIT-03):</td><td></td></tr> </table>	Time 100% Standardization performed (AIT-01):		Time 100% Standardization performed (AIT-03):																	
Time 100% Standardization performed (AIT-01):																					
Time 100% Standardization performed (AIT-03):																					
11.25	- Record the bioreactor temperature and pH. - Prepare sample assembly by flushing 50 mL per SOP-0126420. - Collect approximately 10 mL sample per SOP-0126420. - Perform offline pH check per SOP-0107252. <table border="1"> <tr> <td>Bioreactor Temperature*:</td><td>pH AIT-02:</td><td>pH AIT-04</td></tr> <tr> <td>°C</td><td>(A)</td><td>(B)</td></tr> <tr> <td colspan="2">Offline pH measurement:</td><td>☐ BGA ☐ pH Meter</td></tr> <tr> <td>BGA/ pH Meter ID:</td><td>Sample Temperature*:</td><td>Sample pH (T):</td></tr> <tr> <td></td><td>°C</td><td>(C)</td></tr> <tr> <td colspan="3">*If BGA is used, Bioreactor Temperature and Sample Temperature are the same.</td></tr> </table>	Bioreactor Temperature*:	pH AIT-02:	pH AIT-04	°C	(A)	(B)	Offline pH measurement:		☐ BGA ☐ pH Meter	BGA/ pH Meter ID:	Sample Temperature*:	Sample pH (T):		°C	(C)	*If BGA is used, Bioreactor Temperature and Sample Temperature are the same.				
Bioreactor Temperature*:	pH AIT-02:	pH AIT-04																			
°C	(A)	(B)																			
Offline pH measurement:		☐ BGA ☐ pH Meter																			
BGA/ pH Meter ID:	Sample Temperature*:	Sample pH (T):																			
	°C	(C)																			
*If BGA is used, Bioreactor Temperature and Sample Temperature are the same.																					
11.26	- Calculate the difference between the bioreactor pH and the offline pH reading. - Perform a 1-point standardization if the difference between the bioreactor pH and offline pH reading of either probe is ≤ -0.05 or $\geq +0.05$ pH unit per SOP-0107196. - Record the time the one-point standardization was performed. <table border="1"> <thead> <tr> <th>pH probe</th><th>Bioreactor pH</th><th>Offline pH</th><th>pH Difference</th><th>One-point standardization?</th><th>Time performed</th></tr> </thead> <tbody> <tr> <td><u>AIT-02</u></td><td>_____ (A)</td><td>_____ (C)</td><td>=</td><td>☐ Yes ☐ No</td><td></td></tr> <tr> <td><u>AIT-04</u></td><td>_____ (B)</td><td>_____ (C)</td><td>=</td><td>☐ Yes ☐ No</td><td></td></tr> </tbody> </table>	pH probe	Bioreactor pH	Offline pH	pH Difference	One-point standardization?	Time performed	<u>AIT-02</u>	_____ (A)	_____ (C)	=	☐ Yes ☐ No		<u>AIT-04</u>	_____ (B)	_____ (C)	=	☐ Yes ☐ No			
pH probe	Bioreactor pH	Offline pH	pH Difference	One-point standardization?	Time performed																
<u>AIT-02</u>	_____ (A)	_____ (C)	=	☐ Yes ☐ No																	
<u>AIT-04</u>	_____ (B)	_____ (C)	=	☐ Yes ☐ No																	
11.27	Select the controlling pH and pO₂ probes. <table border="1"> <tr> <td>Controlling pH probe:</td><td>☐ AIT-02 ☐ AIT-04</td></tr> <tr> <td>Controlling pO₂ probe:</td><td>☐ AIT-01 ☐ AIT-03</td></tr> </table>	Controlling pH probe:	☐ AIT-02 ☐ AIT-04	Controlling pO ₂ probe:	☐ AIT-01 ☐ AIT-03																
Controlling pH probe:	☐ AIT-02 ☐ AIT-04																				
Controlling pO ₂ probe:	☐ AIT-01 ☐ AIT-03																				

Step	Description	Performed By/Date	Witnessed By/Date			
11.28	- Progress to the Stabilization phase in the ABEC per SOP-0126420. - Confirm the set points, and recipe below.					
	Growth Stabilize					
	Parameter			Set Point	Status	Allowed Range
	Vessel Temp SP			36.5 °C	Auto	35.0 -38.0
	pH SP			7.35	Auto CO2	6.85 -7.50
	pH deadband			0.10	ON	
	DO SP			100%	OFF	
	Agitation SP			75 rpm	Auto	65 - 85
	Air Overlay SP			5.0 slpm	Auto	1.0 – 10.0
	Primary Air Sparge SP			0.5 slpm	Auto	0.2 – 0.8
	Secondary Air Sparge SP			0	OFF	
	Primary O2 Sparge SP			0	OFF	
	Secondary O2 Sparge SP			0	OFF	

Step	Description	Performed By/Date	Witnessed By/Date																																																																						
11.29	Verify and enable the alarm set points below.																																																																								
	<table><tr><th>Parameter</th><th>Low</th><th>High</th><th>Deviation</th><th>Alarm Status</th></tr><tr><td>Vessel Temp (°C)</td><td>35.0</td><td>38.0</td><td>1.0</td><td>Enable</td></tr><tr><td>Vessel Weight (kg)</td><td>30</td><td>50</td><td>5</td><td>Enable</td></tr><tr><td>Agitation (rpm)</td><td>65</td><td>85</td><td></td><td>Enable</td></tr><tr><td>pH</td><td>6.85</td><td>7.50</td><td>0.11</td><td>Enable</td></tr><tr><td>DO (%)</td><td>10</td><td>110</td><td>10</td><td>Enable</td></tr><tr><td colspan="5">Levitronix 1 Alarms</td></tr><tr><td>Pump Speed (RPM)</td><td>600</td><td>3000</td><td></td><td>Enable</td></tr><tr><td>Flow rate (l/hr)</td><td>2.4</td><td>6.4</td><td></td><td>Enable</td></tr><tr><td>Temp (°C)</td><td></td><td></td><td></td><td>Disable</td></tr><tr><td>TMP (psi)</td><td>0</td><td>0.5</td><td></td><td>Enable</td></tr><tr><td>P1 (psi)</td><td>0</td><td>3</td><td></td><td>Enable</td></tr><tr><td>P2 (psi)</td><td>0</td><td>3</td><td></td><td>Enable</td></tr><tr><td>P3 (psi)</td><td>0</td><td>3</td><td></td><td>Enable</td></tr></table>			Parameter	Low	High	Deviation	Alarm Status	Vessel Temp (°C)	35.0	38.0	1.0	Enable	Vessel Weight (kg)	30	50	5	Enable	Agitation (rpm)	65	85		Enable	pH	6.85	7.50	0.11	Enable	DO (%)	10	110	10	Enable	Levitronix 1 Alarms					Pump Speed (RPM)	600	3000		Enable	Flow rate (l/hr)	2.4	6.4		Enable	Temp (°C)				Disable	TMP (psi)	0	0.5		Enable	P1 (psi)	0	3		Enable	P2 (psi)	0	3		Enable	P3 (psi)	0	3		Enable
	Parameter			Low	High	Deviation	Alarm Status																																																																		
	Vessel Temp (°C)			35.0	38.0	1.0	Enable																																																																		
	Vessel Weight (kg)			30	50	5	Enable																																																																		
	Agitation (rpm)			65	85		Enable																																																																		
	pH			6.85	7.50	0.11	Enable																																																																		
	DO (%)			10	110	10	Enable																																																																		
	Levitronix 1 Alarms																																																																								
	Pump Speed (RPM)			600	3000		Enable																																																																		
	Flow rate (l/hr)			2.4	6.4		Enable																																																																		
	Temp (°C)						Disable																																																																		
	TMP (psi)			0	0.5		Enable																																																																		
	P1 (psi)			0	3		Enable																																																																		
	P2 (psi)			0	3		Enable																																																																		
P3 (psi)	0	3		Enable																																																																					

Step	Description	Performed By/Date	Witnessed By/Date																												
11.30	☐ If inoculation is occurring on day of medium charge, proceed to Step 13.1 and N/A Step 11.31 below. ☐ If inoculation is NOT occurring on day of medium charge, start medium hold. Start medium hold readings in Step 11.31 the day after medium charge. NOTE: Medium may be held at 36.5 °C for 5 days / 120 hours. Inoculation may occur any day during medium hold.																														
11.31	Record the medium hold readings below.																														
	<table border="1"> <thead> <tr> <th>Day</th><th>Actual Temperature (°C) (Set point = 36.5 °C)</th><th>Controlling pO₂ (%) (≥ 85.0%)</th><th>Inoculation (Select Yes or No)</th></tr> </thead> <tbody> <tr> <td rowspan="2">Medium Hold Day 1</td><td rowspan="2"></td><td rowspan="2"></td><td>☐ Yes. Proceed to Step 13.1 and N/A the rest of this table.</td></tr> <tr> <td>☐ No. Continue medium hold.</td></tr> <tr> <td rowspan="2">Medium Hold Day 2</td><td rowspan="2"></td><td rowspan="2"></td><td>☐ Yes. Proceed to Step 13.1 and N/A the rest of this table.</td></tr> <tr> <td>☐ No. Continue medium hold.</td></tr> <tr> <td rowspan="2">Medium Hold Day 3</td><td rowspan="2"></td><td rowspan="2"></td><td>☐ Yes. Proceed to Step 13.1 and N/A the rest of this table.</td></tr> <tr> <td>☐ No. Continue medium hold.</td></tr> <tr> <td rowspan="2">Medium Hold Day 4</td><td rowspan="2"></td><td rowspan="2"></td><td>☐ Yes. Proceed to Step 13.1 and N/A the rest of this table.</td></tr> <tr> <td>☐ No. Continue medium hold.</td></tr> <tr> <td>Medium Hold Day 5</td><td></td><td></td><td>☐ Yes. Proceed to Step 13.1.</td></tr> </tbody> </table>	Day	Actual Temperature (°C) (Set point = 36.5 °C)	Controlling pO ₂ (%) (≥ 85.0%)	Inoculation (Select Yes or No)	Medium Hold Day 1			☐ Yes. Proceed to Step 13.1 and N/A the rest of this table.	☐ No. Continue medium hold.	Medium Hold Day 2			☐ Yes. Proceed to Step 13.1 and N/A the rest of this table.	☐ No. Continue medium hold.	Medium Hold Day 3			☐ Yes. Proceed to Step 13.1 and N/A the rest of this table.	☐ No. Continue medium hold.	Medium Hold Day 4			☐ Yes. Proceed to Step 13.1 and N/A the rest of this table.	☐ No. Continue medium hold.	Medium Hold Day 5			☐ Yes. Proceed to Step 13.1 .		
Day	Actual Temperature (°C) (Set point = 36.5 °C)	Controlling pO ₂ (%) (≥ 85.0%)	Inoculation (Select Yes or No)																												
Medium Hold Day 1			☐ Yes. Proceed to Step 13.1 and N/A the rest of this table.																												
			☐ No. Continue medium hold.																												
Medium Hold Day 2			☐ Yes. Proceed to Step 13.1 and N/A the rest of this table.																												
			☐ No. Continue medium hold.																												
Medium Hold Day 3			☐ Yes. Proceed to Step 13.1 and N/A the rest of this table.																												
			☐ No. Continue medium hold.																												
Medium Hold Day 4			☐ Yes. Proceed to Step 13.1 and N/A the rest of this table.																												
			☐ No. Continue medium hold.																												
Medium Hold Day 5			☐ Yes. Proceed to Step 13.1 .																												

12 Feedcart Set up

Step	Description	Performed By/Date	Witnessed By/Date			
This Section can be completed at any time prior to Section 18 .						
12.1	- Ensure a pre-use cleaning and weight check have been performed on the scale to be used for the permeate waste vessel per SOP-0080224. - Record the Scale ID and calibration expiry.					
	<table border="1"> <tr> <td>Permeate Waste Scale ID</td><td>Calibration Expiry</td></tr> <tr> <td></td><td></td></tr> </table>			Permeate Waste Scale ID	Calibration Expiry	
Permeate Waste Scale ID	Calibration Expiry					
12.2	Obtain the Feed Cart to be used and record the equipment ID below.					
	<table border="1"> <tr> <td>Feed Cart ID</td><td></td></tr> </table>			Feed Cart ID		
Feed Cart ID						
12.3	- Set up the Feed Cart HMI settings per SOP-0126942. - From the Feed Cart HMI, record the Pump ID designated to Feed 5.					
	<table border="1"> <tr> <td>Feed</td><td>Description</td><td>Pump ID</td></tr> <tr> <td>Feed 5</td><td>Permeate</td><td></td></tr> </table>			Feed	Description	Pump ID
Feed	Description	Pump ID				
Feed 5	Permeate					

13 Inoculation Preparation

Step	Description	Performed By/Date	Witnessed By/Date																		
13.1	Verify that the medium in the bioreactor is within the expiry date (Step 11.9).																				
13.2	☐ If a pH measurement / one-point standardization (Step 11.26) has been performed within the same day as the Inoculation process, proceed to Step 13.8 and N/A Steps 13.3 –13.7 . ☐ Otherwise, proceed to Step 13.3 .																				
13.3	- Record the bioreactor temperature and pH. - Prepare sample syringe by flushing 50 mL per SOP-0126420. - Collect approximately 10 mL sample per SOP-0126420. - Perform offline pH check per SOP-0107252. <table border="1"> <tr> <td>Bioreactor Temperature*:</td><td>pH <u>AIT-02</u></td><td>pH <u>AIT-04</u></td></tr> <tr> <td>°C</td><td>(D)</td><td>(E)</td></tr> <tr> <td colspan="2">Offline pH measurement:</td><td>☐ BGA ☐ pH Meter</td></tr> <tr> <td>BGA/ pH Meter ID:</td><td>Sample Temperature*:</td><td>Sample pH (T):</td></tr> <tr> <td></td><td>°C</td><td>(F)</td></tr> <tr> <td colspan="3">*If BGA is used, Bioreactor Temperature and Sample Temperature are the same.</td></tr> </table>	Bioreactor Temperature*:	pH <u>AIT-02</u>	pH <u>AIT-04</u>	°C	(D)	(E)	Offline pH measurement:		☐ BGA ☐ pH Meter	BGA/ pH Meter ID:	Sample Temperature*:	Sample pH (T):		°C	(F)	*If BGA is used, Bioreactor Temperature and Sample Temperature are the same.				
Bioreactor Temperature*:	pH <u>AIT-02</u>	pH <u>AIT-04</u>																			
°C	(D)	(E)																			
Offline pH measurement:		☐ BGA ☐ pH Meter																			
BGA/ pH Meter ID:	Sample Temperature*:	Sample pH (T):																			
	°C	(F)																			
*If BGA is used, Bioreactor Temperature and Sample Temperature are the same.																					
13.4	Calculate the difference between the bioreactor pH and offline pH reading. <table border="1"> <thead> <tr> <th>pH Probe</th><th>Bioreactor pH</th><th></th><th>Offline pH</th><th></th><th>Difference in pH</th></tr> </thead> <tbody> <tr> <td><u>AIT-02</u></td><td>_____ (D)</td><td>–</td><td>_____ (F)</td><td>=</td><td></td></tr> <tr> <td><u>AIT-04</u></td><td>_____ (E)</td><td>–</td><td>_____ (F)</td><td>=</td><td></td></tr> </tbody> </table> - If the pH difference is ≤ -0.10 or $\geq +0.10$ pH unit, collect a second pH sample and repeat the BGA Analysis in Steps 13.5 and 13.6. - If repeat sample is not required, N/A Steps 13.5 and 13.6.	pH Probe	Bioreactor pH		Offline pH		Difference in pH	<u>AIT-02</u>	_____ (D)	–	_____ (F)	=		<u>AIT-04</u>	_____ (E)	–	_____ (F)	=			
pH Probe	Bioreactor pH		Offline pH		Difference in pH																
<u>AIT-02</u>	_____ (D)	–	_____ (F)	=																	
<u>AIT-04</u>	_____ (E)	–	_____ (F)	=																	

Step	Description	Performed By/Date	Witnessed By/Date					
13.5	- Prepare sample assembly by flushing 50 mL per SOP-0126420. - Collect approximately 10 mL sample per SOP-0126420. - Perform offline pH check per SOP-0107252.							
	Bioreactor Temperature*:			pH <u>AIT-02</u>	pH <u>AIT-04</u>			
	°C			(D ₂)	(E ₂)			
	Offline pH measurement:			☐ BGA ☐ pH Meter				
	BGA/ pH Meter ID:			Sample Temperature:	Sample pH (T):			
				°C	(G)			
	*If BGA is used, Bioreactor Temperature and Sample Temperature are the same.							
	13.6			Calculate the difference between the bioreactor pH and offline pH reading.				
pH Probe		Bioreactor pH		Offline pH				Difference in pH
<u>AIT-02</u>		_____ (D)	—	_____ (G)			=	
<u>AIT-04</u>		_____ (E)	—	_____ (G)			=	
If the pH difference is <u>still</u> ≤ -0.10 or ≥ +0.10 pH unit , after repeating the BGA Analysis, contact supervisor/designee for further instructions Record actions taken in Attachment 10: Comments Section .								
Supervisor/designee contacted: _____								
13.7	- Perform a 1-point standardization on the bioreactor transmitters if the difference between the bioreactor pH and offline pH reading is ≤ -0.05 or ≥ +0.05 pH unit per SOP-0126420. NOTE: Probes are standardized as needed independently of each other. - Record the time the one-point standardization was performed.							
	pH probe			One-point standardization performed?	Time performed:			
	<u>AIT-02</u>			☐ Yes ☐ No				
	<u>AIT-04</u>			☐ Yes ☐ No				

Step	Description	Performed By/Date	Witnessed By/Date	
13.8	<u>Seed Medium Sample:</u> - Obtain a Sampling Bag Assembly (250 mL) (6000391). - Record assembly and welder information. - Label this assembly with "INT-034", 8012420 and batch number, bioreactor ID, "Pre-Inoculation 100 L Seed Medium", initials, and date. - Weld the assembly to the sample port per SOP-0107187 or SOP-0107297. - Record the sample collection in Attachment 7: Retain Samples.			
	Welder ID			
	Part No.:			
	Batch No.:			
	Exp. Date:			
	SAP Consumption By/Date			
13.9	- Dispense approximately 100 mL to be stored as investigational sample at 2 - 8 °C. - Seal off assembly for removal and record sealer information.			
	Sealer ID			
13.10	- Allow pO₂ to stabilize. - Record the initial pO₂ value from the controlling probe selected in Step 11.27. NOTE: If the initial pO₂ recorded is < 85.0%, contact supervisor/designee and record the instructions provided in Attachment 10: Comments Section. Supervisor/designee contacted: _____	pO ₂ :		
		_____ % (≥ 85.0%)		

Step	Description	Performed By/Date	Witnessed By/Date			
13.11	- Initiate the Inoculate/Grow 1 recipe in the ABEC HMI per SOP-0126420. - Verify the setpoints below:					
	Grow 1 (N-2)					
	Parameter			Set point	Status	Allowed Range
	Vessel Temp SP			36.5 °C	Auto	35.0 -38.0
	pH SP			7.35	Auto CO2	6.85 -7.50
	pH deadband			0.10		
	DO SP			50%		10% -110%
	Agitation SP			75 rpm	Cascade	60 -150
	Air Overlay SP			5.0 slpm	Auto	1.0 – 10.0
	Primary Air Sparge SP			0.5 slpm	Auto	0.2 – 0.8
	Secondary Air Sparge SP			0	Off	
	Primary O2 Sparge SP			0	Cascade	
	Secondary O2 Sparge SP			0	Off	
	pH Strategy				Auto CO2	
	DO Strategy				Profile	
13.12	- Ensure the permeate clamp on TFF 1 is closed. - Verify the Levitronix 1 settings below:					
	Flow Rate			Control Selection		
	4.4 lpm			Flow		

14 Inoculation

Step	Description	Performed By/Date	Witnessed By/Date								
14.1	- Due to initiating the Grow 1 recipe prior to inoculation, alarms may be triggered. Monitor accordingly between step 14.2 - 14.8. - If parameters exceed their allowable range per Step 13.11, contact supervisor/designee and record actions taken in Attachment 10: Comments Section.										
14.2	Before inoculating, record the 100 L bioreactor start weight from the Loops screen. NOTE: This weight range does not include the TFF Loop. If the bioreactor start weight is not within the specification, contact the supervisor/designee and record action taken in Attachment 10: Comments Section.	Start Weight: _____ kg (35 –45 kg) Supervisor/designee contacted: _____									
14.3	Obtain the following material and record information below.										
	<table border="1"> <thead> <tr> <th>Description</th><th>Part No.</th><th>Batch No.</th><th>Exp. Date</th></tr> </thead> <tbody> <tr> <td>1/4" Pump Runner with AQS</td><td>6005502</td><td></td><td></td></tr> </tbody> </table>	Description	Part No.	Batch No.	Exp. Date	1/4" Pump Runner with AQS	6005502				
Description	Part No.	Batch No.	Exp. Date								
1/4" Pump Runner with AQS	6005502										
	SAP Consumption By/Date										
14.4	- Obtain the "Shake Flask Inoculum/Transfer" Vessel labeled with PN: 8012419. - Record the inoculum culture part and batch number.										
	<table border="1"> <thead> <tr> <th>Inoculum Part No.:</th><th>Inoculum Batch No.:</th></tr> </thead> <tbody> <tr> <td></td><td></td></tr> </tbody> </table>	Inoculum Part No.:	Inoculum Batch No.:								
Inoculum Part No.:	Inoculum Batch No.:										
14.5	- Weld the "Shake Flask Inoculum/Transfer" Vessel to the pump runner per SOP-0080257. - Connect the pump runner to the appropriate inoculation line per SOP-0126420.										
	<table border="1"> <tbody> <tr> <td>Welder ID:</td><td></td></tr> </tbody> </table>	Welder ID:									
Welder ID:											

Step	Description				Performed By/Date	Witnessed By/Date	
14.6	- Inoculate the 100 L bioreactor by transferring the entire contents of the “Shake Flask Inoculum/Transfer” Vessel per SOP-0126420. - Record the weight of the 100 L bioreactor after the inoculation transfer from the Loops screen.						
	Inoculum Addition Start Time:		Inoculum Addition End Time:				Bioreactor Weight After Inoculation:
							kg
	SAP Consumption performed by/date						
14.7	Remove the Inoculum Transfer Vessel from the bioreactor per SOP-0080812 or SOP-0080522.						
	Sealer ID:						
14.8	Take a VCD sample from the bioreactor record the initial VCD below per SOP-0126420.						
	VCD (≥ 0.2 x 10^6)		_____ x 10^6 cells/mL				
14.9	On the Recipe Alarms page, change the Interlock alarm per the table below:						
	Bioreactor System Alarm		Low Interlock				
	Vessel Weight (kg)		20				
14.10	- Verify the alarm set points for Inoculate/Grow 1 (N-2) below. - Verify alarms are enabled.						
	Parameter	Low	High	Deviation			Alarm Status
	Vessel Temp (°C)	35.0	38.0	1.0			Enable
	Vessel Weight (kg)	35	55	5			Enable
	Agitation (rpm)	60	150				Enable
	pH	6.85	7.50	0.11			Enable
	DO (%)	10	110	10			Enable

15 Daily Sampling – N-2 Seed Phase

Step	Description	Performed By/Date	Witnessed By/Date
15.1	On a daily basis, including Day 0 or as instructed by supervisor/designee: • Prepare sample assembly by flushing 50 mL per SOP-0126420. • Collect approximately 10 mL sample from the bioreactor per SOP-0126420 and perform the following analysis: – Perform offline pH check per SOP-0107252. – Perform a one-point correction of pH probes per SOP-0126420 when required. NOTE: If one-point correction is required, it must be performed prior to any additions. – Perform offline osmolality check per SOP-0080509. – Measure cell density and percent viability per SOP-0115565. NOTE: Use CHO Method 2 for cell count with Vi-CELL BLU. • Perform glucose and lactate analysis. • Record bioreactor conditions in Table 1: Bioreactor Conditions N-2. • If needed, additional sample pages can be found in Attachment 8: Additional Sampling Pages (N-2).		

Table 1: Bioreactor Conditions N-2

Perform daily sampling of the bioreactor.			BGA ID:	Measurements	Result	
Day	Seed Day 0			pH (T) (X.XX)		
Time				pCO ₂ (T) (X.X)	mmHg	
Select Primary DO Probe	☐ AIT-01 / ☐ AIT-03			pO ₂ (T) (X.X)	mmHg	
Bioreactor Temp. (°C) (XX.X) (35.0 - 38.0°C)				Bioreactor Glu (X)	mg/dL	
Pressure (psig) (X.XX) (0 – 0.6 psig)						
pO ₂ (%) (X.X) (10.0 - 110.0%)			Lac (mmol/L) (X.X)	Lactate Dilution	Calculated Lactate (mmol/L) (X.X)	
Bioreactor pH (X.XX) (6.85 - 7.50)				☐ N/A	☐ N/A	
Agitation (rpm) (X) (60 – 150 rpm)			Adjustment <i>Perform a one-point calibration if pH difference is < -0.05 or > 0.05. Take a 2nd pH sample if pH difference is < -0.10 or > 0.10. Record information from both probes. Only primary probe results will be used for processing decisions. The secondary probe results are for information only (FIO).</i>			
Overlay Air (slpm) (X.X) (1.0 – 10.0 slpm)			Select Primary pH Probe	Bioreactor pH (X.XX)	Offline pH (T) (X.XX)	pH Difference
Air Sparge (slpm) (X.X) (0.2 - 0.8 slpm)			☐ AIT-02			
			☐ AIT-04			
O ₂ Sparge (slpm) (X.X)			Is one-point calibration required based on 1 st pH sample results?			☐ Yes ☐ No
CO ₂ Sparge (slpm) (X.X)			Is a 2 nd pH sample required? If yes, go to Attachment 6: Additional pH Monitoring – pH samples' information.			☐ Yes ☐ No
Bioreactor Weight (kg) (X)			Pipettor ID (if applicable)	Measurements	Results	
Foam Check: Is Antifoam addition Needed? Proceed to Attachment 4 for manual addition.				Vi-Cell Dilution (X)		
			Vi-Cell ID:	VCD (X.X)	x 10 ⁶ cells/mL	
☐ Yes	☐ No			Viability (X)	%	
Was any pump tubing shifted?	☐ Yes ☐ N/A		Osmometer ID	Osmolality (X)		
			Has Dilution criteria been met? (VCD ≥ 2.1 x 10 ⁶ cells/mL) Continue to Section 16 for Medium Dilution and N/A all subsequent sample pages			☐ Yes ☐ No
Performed By/Date:				Witness By/Date:		

Table 1: Bioreactor Conditions N-2

Perform daily sampling of the bioreactor.			BGA ID:	Measurements	Result	
Day	Seed Day 1			pH (T) (X.XX)		
Time				pCO ₂ (T) (X.X)	mmHg	
Select Primary DO Probe	☐ AIT-01 / ☐ AIT-03			pO ₂ (T) (X.X)	mmHg	
Bioreactor Temp. (°C) (XX.X) (35.0 - 38.0°C)				Bioreactor Glu (X)	mg/dL	
Pressure (psig) (X.XX) (0 – 0.6 psig)						
pO ₂ (%) (X.X) (10.0 - 110.0%)			Lac (mmol/L) (X.X)	Lactate Dilution	Calculated Lactate (mmol/L) (X.X)	
Bioreactor pH (X.XX) (6.85 - 7.50)				☐ N/A	☐ N/A	
Agitation (rpm) (X) (60 – 150 rpm)			Adjustment Perform a one-point calibration if pH difference is < -0.05 or > 0.05. Take a 2nd pH sample if pH difference is < -0.10 or > 0.10. Record information from both probes. Only primary probe results will be used for processing decisions. The secondary probe results are for information only (FIO).			
Overlay Air (slpm) (X.X) (1.0 – 10.0 slpm)			Select Primary pH Probe	Bioreactor pH (X.XX)	Offline pH (T) (X.XX)	pH Difference
Air Sparge (slpm) (X.X) (0.2 - 0.8 slpm)			☐ AIT-02			
			☐ AIT-04			
O ₂ Sparge (slpm) (X.X)			Is one-point calibration required based on 1 st pH sample results?			☐ Yes ☐ No
CO ₂ Sparge (slpm) (X.X)			Is a 2 nd pH sample required? If yes, go to Attachment 6: Additional pH Monitoring – pH samples' information.			☐ Yes ☐ No
Bioreactor Weight (kg) (X)			Pipettor ID (if applicable)	Measurements	Results	
Foam Check: Is Antifoam addition Needed? Proceed to Attachment 4 for manual addition.				Vi-Cell Dilution (X)		
			Vi-Cell ID:	VCD (X.X)	x 10 ⁶ cells/mL	
☐ Yes	☐ No			Viability (X)	%	
Was any pump tubing shifted?	☐ Yes ☐ N/A		Osmometer ID	Osmolality (X)		
			Has Dilution criteria been met? (VCD ≥ 2.1 x 10 ⁶ cells/mL) Continue to Section 16 for Medium Dilution and N/A all subsequent sample pages			☐ Yes ☐ No
Performed By/Date:			Witness By/Date:			

Table 1: Bioreactor Conditions N-2

Perform daily sampling of the bioreactor.			BGA ID:	Measurements	Result	
Day	Seed Day 2			pH (T) (X.XX)		
Time				pCO ₂ (T) (X.X)	mmHg	
Select Primary DO Probe	☐ AIT-01 / ☐ AIT-03			pO ₂ (T) (X.X)	mmHg	
Bioreactor Temp. (°C) (XX.X) (35.0 - 38.0°C)				Bioreactor Glu (X)	mg/dL	
Pressure (psig) (X.XX) (0 – 0.6 psig)						
pO ₂ (%) (X.X) (10.0 - 110.0%)			Lac (mmol/L) (X.X)	Lactate Dilution	Calculated Lactate (mmol/L) (X.X)	
Bioreactor pH (X.XX) (6.85 - 7.50)				☐ N/A	☐ N/A	
Agitation (rpm) (X) (60 – 150 rpm)			Adjustment Perform a one-point calibration if pH difference is < -0.05 or > 0.05. Take a 2nd pH sample if pH difference is < -0.10 or > 0.10. Record information from both probes. Only primary probe results will be used for processing decisions. The secondary probe results are for information only (FIO).			
Overlay Air (slpm) (X.X) (1.0 – 10.0 slpm)			Select Primary pH Probe	Bioreactor pH (X.XX)	Offline pH (T) (X.XX)	pH Difference
Air Sparge (slpm) (X.X) (0.2 - 0.8 slpm)			☐ AIT-02			
			☐ AIT-04			
O ₂ Sparge (slpm) (X.X)			Is one-point calibration required based on 1 st pH sample results?			☐ Yes ☐ No
CO ₂ Sparge (slpm) (X.X)			Is a 2 nd pH sample required? If yes, go to Attachment 6: Additional pH Monitoring – pH samples' information.			☐ Yes ☐ No
Bioreactor Weight (kg) (X)			Pipettor ID (if applicable)	Measurements	Results	
Foam Check: Is Antifoam addition Needed? Proceed to Attachment 4 for manual addition.				Vi-Cell Dilution (X)		
			Vi-Cell ID:	VCD (X.X)	x 10 ⁶ cells/mL	
☐ Yes	☐ No			Viability (X)	%	
Was any pump tubing shifted?	☐ Yes ☐ N/A		Osmometer ID	Osmolality (X)		
			Has Dilution criteria been met? (VCD ≥ 2.1 x 10 ⁶ cells/mL) Continue to Section 16 for Medium Dilution and N/A all subsequent sample pages			☐ Yes ☐ No
Performed By/Date:			Witness By/Date:			

Table 1: Bioreactor Conditions N-2

Perform daily sampling of the bioreactor.			BGA ID:	Measurements	Result	
Day	Seed Day 3			pH (T) (X.XX)		
Time				pCO ₂ (T) (X.X)	mmHg	
Select Primary DO Probe	☐ AIT-01 / ☐ AIT-03			pO ₂ (T) (X.X)	mmHg	
Bioreactor Temp. (°C) (XX.X) (35.0 - 38.0°C)				Bioreactor Glu (X)	mg/dL	
Pressure (psig) (X.XX) (0 - 0.6 psig)				Lactate Dilution	Calculated Lactate (mmol/L) (X.X)	
pO ₂ (%) (X.X) (10.0 - 110.0%)			Lac (mmol/L) (X.X)			
Bioreactor pH (X.XX) (6.85 - 7.50)				☐ N/A	☐ N/A	
Agitation (rpm) (X) (60 - 150 rpm)			Adjustment Perform a one-point calibration if pH difference is < -0.05 or > 0.05. Take a 2nd pH sample if pH difference is < -0.10 or > 0.10. Record information from both probes. Only primary probe results will be used for processing decisions. The secondary probe results are for information only (FIO).			
Overlay Air (slpm) (X.X) (1.0 - 10.0 slpm)			Select Primary pH Probe	Bioreactor pH (X.XX)	Offline pH (T) (X.XX)	pH Difference
Air Sparge (slpm) (X.X) (0.2 - 0.8 slpm)			☐ AIT-02			
			☐ AIT-04			
O ₂ Sparge (slpm) (X.X)			Is one-point calibration required based on 1 st pH sample results?			☐ Yes ☐ No
CO ₂ Sparge (slpm) (X.X)			Is a 2 nd pH sample required? If yes, go to Attachment 6: Additional pH Monitoring – pH samples' information.			☐ Yes ☐ No
Bioreactor Weight (kg) (X)			Pipettor ID (if applicable)	Measurements	Results	
Foam Check: Is Antifoam addition Needed? Proceed to Attachment 4 for manual addition.				Vi-Cell Dilution (X)		
			Vi-Cell ID:	VCD (X.X)	x 10 ⁶ cells/mL	
☐ Yes	☐ No			Viability (X)	%	
Was any pump tubing shifted?	☐ Yes ☐ N/A		Osmometer ID	Osmolality (X)		
			Has Dilution criteria been met? (VCD ≥ 2.1 x 10 ⁶ cells/mL) Continue to Section 16 for Medium Dilution and N/A all subsequent sample pages			☐ Yes ☐ No
Performed By/Date:			Witness By/Date:			

Table 1: Bioreactor Conditions N-2

Perform daily sampling of the bioreactor.			BGA ID:	Measurements	Result	
Day	Seed Day 4			pH (T) (X.XX)		
Time				pCO ₂ (T) (X.X)	mmHg	
Select Primary DO Probe	☐ AIT-01 / ☐ AIT-03			pO ₂ (T) (X.X)	mmHg	
Bioreactor Temp. (°C) (XX.X) (35.0 - 38.0°C)				Bioreactor Glu (X)	mg/dL	
Pressure (psig) (X.XX) (0 – 0.6 psig)						
pO ₂ (%) (X.X) (10.0 - 110.0%)			Lac (mmol/L) (X.X)	Lactate Dilution	Calculated Lactate (mmol/L) (X.X)	
Bioreactor pH (X.XX) (6.85 - 7.50)				☐ N/A	☐ N/A	
Agitation (rpm) (X) (60 – 150 rpm)			Adjustment <i>Perform a one-point calibration if pH difference is < -0.05 or > 0.05. Take a 2nd pH sample if pH difference is < -0.10 or > 0.10. Record information from both probes. Only primary probe results will be used for processing decisions. The secondary probe results are for information only (FIO).</i>			
Overlay Air (slpm) (X.X) (1.0 – 10.0 slpm)			Select Primary pH Probe	Bioreactor pH (X.XX)	Offline pH (T) (X.XX)	pH Difference
Air Sparge (slpm) (X.X) (0.2 - 0.8 slpm)			☐ AIT-02			
			☐ AIT-04			
O ₂ Sparge (slpm) (X.X)			Is one-point calibration required based on 1 st pH sample results?			☐ Yes ☐ No
CO ₂ Sparge (slpm) (X.X)			Is a 2 nd pH sample required? If yes, go to Attachment 6: Additional pH Monitoring – pH samples' information.			☐ Yes ☐ No
Bioreactor Weight (kg) (X)			Pipettor ID (if applicable)	Measurements	Results	
Foam Check: Is Antifoam addition Needed? Proceed to Attachment 4 for manual addition.				Vi-Cell Dilution (X)		
			Vi-Cell ID:	VCD (X.X)	x 10 ⁶ cells/mL	
☐ Yes	☐ No			Viability (X)	%	
Was any pump tubing shifted?	☐ Yes ☐ N/A		Osmometer ID	Osmolality (X)		
			Has Dilution criteria been met? (VCD ≥ 2.1 x 10 ⁶ cells/mL) Continue to Section 16 for Medium Dilution and N/A all subsequent sample pages			☐ Yes ☐ No
Performed By/Date:				Witness By/Date:		

16 Medium Dilution

Step	Description	Performed By/Date	Witnessed By/Date		
16.1	When the viable cell density is $\geq 2.1 \times 10^6$ cells/mL, prepare to add medium. NOTE: Use the CHO Method 2 for cell count with Vi-CELL BLU.				
	Seed Day:			Viable Cell Density (VCD):	Viability:
				_____ x 10^6 cells/mL (2.1 – 4.4 x 10^6 cells/mL)	_____ % ($\geq 90\%$)
	Contact Supervisor if: - the viability is less than 90% and proceed with comment. - VCD is not met by Seed Day 4. - Additional samples can be taken under supervisor discretion. Document daily sample results in Attachment 8: Additional Sampling Pages (N-2). NOTE: Maximum time allowed in N-2 step is 120 hours. - VCD is greater than 4.4×10^6 cells/mL. Supervisor/designee contacted: _____				
16.2	<u>Investigational Sample:</u>				
	- Obtain a sampling assembly (6000391). - Record assembly and welder information. - Label this assembly with “INT-034”, 8012420 and batch number, bioreactor ID, “100 L Bioreactor Pre-Medium Dilution”, initials, and date. - Weld the assembly to the sample port per SOP-0107187 or SOP-0107297. - Record retain sample per Attachment 7: Retain Samples.				
	Welder ID				
	Part No.:				
	Batch No.:				
	Exp. Date:				
	SAP Consumption By/Date				

Step	Description	Performed By/Date	Witnessed By/Date								
16.3	- Dispense approximately 50 mL to be stored as investigational sample. - Seal off assembly for removal and record sealer information.										
	Sealer ID										
16.4	Obtain the following materials and record the information below.										
	<table border="1"> <thead> <tr> <th>Description</th><th>Part No.</th><th>Batch No.</th><th>Expiry Date</th></tr> </thead> <tbody> <tr> <td>Male Purefit to AQS</td><td>6005484*</td><td></td><td></td></tr> </tbody> </table>			Description	Part No.	Batch No.	Expiry Date	Male Purefit to AQS	6005484*		
	Description			Part No.	Batch No.	Expiry Date					
	Male Purefit to AQS			6005484*							
	*If PN 6005484 is not available, weld the CCMB1 Medium in Step 16.5 .										
SAP Consumption Performed By/Date											
16.5	- Obtain CCMB1 Medium (M-030-V38) (PN: 8010764). - Connect medium (PN: 8010764) to the DC charge line per SOP-0162420.										
	Part Number										
	Batch Number										
	Exp. Date										
	Welder ID, if applicable										

Step	Description	Performed By/Date	Witnessed By/Date
16.6	<u>Dilution Medium Sample:</u> - Obtain a sampling assembly (6000391). - Record the assembly and welder information. - Label this assembly with “INT-034”, 8012420 and batch number, bioreactor ID, “100 L Dilution Medium”, initials, and date. - Weld the assembly to the Medium bag per SOP-0107187 or SOP-0107297.		
	Welder ID		
	Part No.:		
	Batch No.:		
	Exp. Date:		
	SAP Consumption Performed By/Date		
16.7	- Dispense approximately 50 mL of medium into the sample bag to be stored as investigational sample at 2 - 8 °C. - Record retain sample per Attachment 7: Retain Samples. - Seal off assembly for removal and record sealer information.		
	Sealer ID		

Step	Description	Performed By/Date	Witnessed By/Date			
16.8	- Initiate the Grow 2 recipe in the ABEC HMI per SOP-0126420. - Verify the setpoints below:					
	Grow 2 (N-1)					
	Parameter			Set point	Status	Allowed Range
	Vessel Temp SP			36.5 °C	Auto	35.0 -38.0
	pH SP			7.10	Auto CO2	6.70 -7.50
	pH deadband			0.20		
	DO SP			50%		10% -110%
	Agitation SP			75 rpm	Cascade	60 -150
	Air Overlay SP			5.0 slpm	Auto	1.0 – 10.0
	Primary Air Sparge SP			0.5 slpm	Auto	0.5 – 5.0
	Secondary Air Sparge SP			0	Off	
	Primary O2 Sparge SP			0	Cascade	
	Secondary O2 Sparge SP			0	Off	
	pH Strategy				Auto CO2	
	DO Strategy				Profile	
	16.9			- Ensure the permeate clamp on TFF 1 is closed. - Verify the Levitronix 1 settings below:		
Flow Rate		Control Selection				
4.4 lpm		Flow				

Step	Description	Performed By/Date	Witnessed By/Date									
16.10	- Prime the medium addition line and record the weight of the bioreactor before medium transfer. - Calculate the target weight of the bioreactor after medium dilution.											
	<table border="1"> <tr> <td>Current Bioreactor Weight from HMI</td><td rowspan="2">+</td><td>Target Weight of Medium to Add</td><td rowspan="2">=</td><td>Target Bioreactor Weight After Medium Dilution</td></tr> <tr> <td>_____ kg</td><td>55 kg</td><td>_____ kg</td></tr> </table>			Current Bioreactor Weight from HMI	+	Target Weight of Medium to Add	=	Target Bioreactor Weight After Medium Dilution	_____ kg	55 kg	_____ kg	
	Current Bioreactor Weight from HMI			+		Target Weight of Medium to Add		=	Target Bioreactor Weight After Medium Dilution			
_____ kg	55 kg	_____ kg										
16.11	Slowly transfer 55 ± 2.75 kg of medium into the bioreactor per SOP-0126420 until the target weight is achieved.	Bioreactor Weight After Medium Dilution										
		_____ kg										
16.12	Record the end of media dilution time below.											
	End Time of Media Dilution											
16.13	Calculate the actual amount of medium transferred to the bioreactor: NOTE: Ensure that the amount transferred to the bioreactor is within the specified range.											
	<table border="1"> <tr> <td>Bioreactor Weight After Medium Dilution</td><td rowspan="2">-</td><td>Bioreactor Weight Before Medium Dilution</td><td rowspan="2">=</td><td>Amount of Medium Transferred</td></tr> <tr> <td>_____ kg</td><td>_____ kg</td><td>_____ kg (52.25 – 57.75 kg)</td></tr> </table>	Bioreactor Weight After Medium Dilution		-	Bioreactor Weight Before Medium Dilution	=	Amount of Medium Transferred	_____ kg	_____ kg	_____ kg (52.25 – 57.75 kg)		
	Bioreactor Weight After Medium Dilution	-			Bioreactor Weight Before Medium Dilution		=	Amount of Medium Transferred				
	_____ kg			_____ kg	_____ kg (52.25 – 57.75 kg)							
	SAP Consumption Performed By/Date											
16.14	Calculate the TCCV after media dilution.											
	<table border="1"> <tr> <td>Bioreactor Weight After Medium Dilution (Step 16.11)</td><td rowspan="2">+</td><td>TFF 1 Hold up Volume (Step 11.14)</td><td rowspan="2">=</td><td>TCCV (Working Volume)</td></tr> <tr> <td>_____</td><td>_____</td><td>_____</td></tr> </table>	Bioreactor Weight After Medium Dilution (Step 16.11)		+	TFF 1 Hold up Volume (Step 11.14)	=	TCCV (Working Volume)	_____	_____	_____		
	Bioreactor Weight After Medium Dilution (Step 16.11)	+			TFF 1 Hold up Volume (Step 11.14)		=	TCCV (Working Volume)				
_____	_____		_____									
16.15	On the Recipe Alarms page, change the Interlock alarm per the table below:											
	Bioreactor System Alarm	Low Interlock										
	Vessel Weight (kg)	65										

Step	Description	Performed By/Date	Witnessed By/Date																														
16.16	- Verify the alarm set points for Grow 2 (N-1) below. - Verify alarms are enabled.																																
	<table><tr><th>Parameter</th><th>Low</th><th>High</th><th>Deviation</th><th>Alarm Status</th></tr><tr><td>Vessel Temp (°C)</td><td>35.0</td><td>38.0</td><td>1.0</td><td>Enable</td></tr><tr><td>Vessel Weight (kg)</td><td>85</td><td>115</td><td>10</td><td>Enable</td></tr><tr><td>Agitation (rpm)</td><td>60</td><td>150</td><td></td><td>Enable</td></tr><tr><td>pH</td><td>6.70</td><td>7.50</td><td>0.22</td><td>Enable</td></tr><tr><td>DO (%)</td><td>10</td><td>110</td><td>10</td><td>Enable</td></tr></table>			Parameter	Low	High	Deviation	Alarm Status	Vessel Temp (°C)	35.0	38.0	1.0	Enable	Vessel Weight (kg)	85	115	10	Enable	Agitation (rpm)	60	150		Enable	pH	6.70	7.50	0.22	Enable	DO (%)	10	110	10	Enable
	Parameter			Low	High	Deviation	Alarm Status																										
	Vessel Temp (°C)			35.0	38.0	1.0	Enable																										
	Vessel Weight (kg)			85	115	10	Enable																										
	Agitation (rpm)			60	150		Enable																										
	pH			6.70	7.50	0.22	Enable																										
	DO (%)			10	110	10	Enable																										
16.17	Proceed to Section 17 to begin N-1 Seed Phase Sampling on day of medium dilution.																																

17 Daily Sampling – N-1 Seed Phase

Step	Description	Performed By/Date	Witnessed By/Date
17.1	On a daily basis, including Day 0 or as instructed by supervisor/designee: • Prepare sample assembly by flushing 50 mL per SOP-0126420. • Collect approximately 10 mL sample from the bioreactor per SOP-0126420 and perform the following analysis: – Perform offline pH check per SOP-0107252. – Perform a one-point correction of pH probes per SOP-0126420 when required. NOTE: If one-point correction is required, it must be performed prior to any additions. – Perform offline osmolality check per SOP-0080509. – Measure cell density and percent viability per SOP-0115565. NOTE: Use CHO Method 2 for cell count with Vi-CELL BLU. • Perform glucose and lactate analysis. • Record bioreactor conditions in Table 2: Bioreactor Conditions N-1. • If needed, additional sample pages can be found in Attachment 9: Additional Sampling Pages (N-1).		

Table 2: Bioreactor Conditions N-1

Perform daily sampling of the bioreactor.			BGA ID:	Measurements	Result	
Day	Seed Day _____			pH (T) (X.XX)		
Time				pCO ₂ (T) (X.X)	mmHg	
Select Primary DO Probe	☐ AIT-01 / ☐ AIT-03			pO ₂ (T) (X.X)	mmHg	
Bioreactor Temp. (°C) (XX.X) (35.0 - 38.0°C)				Bioreactor Glu (X)	mg/dL	
Pressure (psig) (X.XX) (0 – 0.6 psig)						
pO ₂ (%) (X.X) (10.0 - 110.0%)			Lac (mmol/L) (X.X)	Lactate Dilution	Calculated Lactate (mmol/L) (X.X)	
Bioreactor pH (X.XX) (6.70 - 7.50)				☐ N/A	☐ N/A	
Agitation (rpm) (X) (60 – 150 rpm)			Adjustment <i>Perform a one-point calibration if pH difference is < -0.05 or > 0.05. Take a 2nd pH sample if pH difference is < -0.10 or > 0.10. Record information from both probes. Only primary probe results will be used for processing decisions. The secondary probe results are for information only (FIO).</i>			
Overlay Air (slpm) (X.X) (1.0 – 10.0 slpm)			Select Primary pH Probe	Bioreactor pH (X.XX)	Offline pH (T) (X.XX)	pH Difference
Air Sparge (slpm) (X.X) (0.5 - 5.0 slpm)			☐ AIT-02			
			☐ AIT-04			
O ₂ Sparge (slpm) (X.X)			Is one-point calibration required based on 1 st pH sample results?			☐ Yes ☐ No
CO ₂ Sparge (slpm) (X.X)			Is a 2 nd pH sample required? If yes, go to Attachment 6: Additional pH Monitoring – pH samples' information.			☐ Yes ☐ No
Bioreactor Weight (kg) (X)			Pipettor ID (if applicable)	Measurements	Results	
Foam Check: Is Antifoam addition Needed? Proceed to Attachment 4 for manual addition.				Vi-Cell Dilution (X)		
			Vi-Cell ID:	VCD (X.X)	x 10 ⁶ cells/mL	
☐ Yes	☐ No			Viability (X)	%	
Was any pump tubing shifted?	☐ Yes ☐ N/A		Osmometer ID	Osmolality (X)		
			Has Perfusion criteria been met? (VCD ≥ 1.8 x 10 ⁶ cells/mL AND 48 ± 6 hours post Medium Dilution) Continue to Section 18 for Perfusion Initiation.			☐ Yes ☐ No
Performed By/Date:			Witness By/Date:			

Table 2: Bioreactor Conditions N-1

Perform daily sampling of the bioreactor.			BGA ID:	Measurements	Result	
Day	Seed Day _____			pH (T) (X.XX)		
Time				pCO ₂ (T) (X.X)	mmHg	
Select Primary DO Probe	☐ AIT-01 / ☐ AIT-03			pO ₂ (T) (X.X)	mmHg	
Bioreactor Temp. (°C) (XX.X) (35.0 - 38.0°C)				Bioreactor Glu (X)	mg/dL	
Pressure (psig) (X.XX) (0 - 0.6 psig)						
pO ₂ (%) (X.X) (10.0 - 110.0%)			Lac (mmol/L) (X.X)	Lactate Dilution	Calculated Lactate (mmol/L) (X.X)	
Bioreactor pH (X.XX) (6.70 - 7.50)				☐ N/A	☐ N/A	
Agitation (rpm) (X) (60 - 150 rpm)			Adjustment Perform a one-point calibration if pH difference is < -0.05 or > 0.05. Take a 2nd pH sample if pH difference is < -0.10 or > 0.10. Record information from both probes. Only primary probe results will be used for processing decisions. The secondary probe results are for information only (FIO).			
Overlay Air (slpm) (X.X) (1.0 - 10.0 slpm)			Select Primary pH Probe	Bioreactor pH (X.XX)	Offline pH (T) (X.XX)	pH Difference
Air Sparge (slpm) (X.X) (0.5 - 5.0 slpm)			☐ AIT-02			
			☐ AIT-04			
O ₂ Sparge (slpm) (X.X)			Is one-point calibration required based on 1 st pH sample results?			☐ Yes ☐ No
CO ₂ Sparge (slpm) (X.X)			Is a 2 nd pH sample required? If yes, go to Attachment 6: Additional pH Monitoring – pH samples' information.			☐ Yes ☐ No
Bioreactor Weight (kg) (X)			Pipettor ID (if applicable)	Measurements	Results	
Foam Check: Is Antifoam addition Needed? Proceed to Attachment 4 for manual addition.				Vi-Cell Dilution (X)		
			Vi-Cell ID:	VCD (X.X)	x 10 ⁶ cells/mL	
☐ Yes	☐ No			Viability (X)	%	
Was any pump tubing shifted?	☐ Yes ☐ N/A		Osmometer ID	Osmolality (X)		
			Has Perfusion criteria been met? (VCD ≥ 1.8 x 10 ⁶ cells/mL AND 48 ± 6 hours post Medium Dilution) Continue to Section 18 for Perfusion Initiation.			☐ Yes ☐ No
Performed By/Date:			Witness By/Date:			

Table 2: Bioreactor Conditions N-1

Perform daily sampling of the bioreactor.			BGA ID:	Measurements	Result	
Day	Seed Day _____			pH (T) (X.XX)		
Time				pCO ₂ (T) (X.X)	mmHg	
Select Primary DO Probe	☐ AIT-01 / ☐ AIT-03			pO ₂ (T) (X.X)	mmHg	
Bioreactor Temp. (°C) (XX.X) (35.0 - 38.0°C)				Bioreactor Glu (X)	mg/dL	
Pressure (psig) (X.XX) (0 - 0.6 psig)						
pO ₂ (%) (X.X) (10.0 - 110.0%)			Lac (mmol/L) (X.X)	Lactate Dilution	Calculated Lactate (mmol/L) (X.X)	
Bioreactor pH (X.XX) (6.70 - 7.50)				☐ N/A	☐ N/A	
Agitation (rpm) (X) (60 - 150 rpm)			Adjustment Perform a one-point calibration if pH difference is < -0.05 or > 0.05. Take a 2nd pH sample if pH difference is < -0.10 or > 0.10. Record information from both probes. Only primary probe results will be used for processing decisions. The secondary probe results are for information only (FIO).			
Overlay Air (slpm) (X.X) (1.0 - 10.0 slpm)			Select Primary pH Probe	Bioreactor pH (X.XX)	Offline pH (T) (X.XX)	pH Difference
Air Sparge (slpm) (X.X) (0.5 - 5.0 slpm)			☐ AIT-02			
			☐ AIT-04			
O ₂ Sparge (slpm) (X.X)			Is one-point calibration required based on 1 st pH sample results?			☐ Yes ☐ No
CO ₂ Sparge (slpm) (X.X)			Is a 2 nd pH sample required? If yes, go to Attachment 6: Additional pH Monitoring – pH samples' information.			☐ Yes ☐ No
Bioreactor Weight (kg) (X)			Pipettor ID (if applicable)	Measurements	Results	
Foam Check: Is Antifoam addition Needed? Proceed to Attachment 4 for manual addition.				Vi-Cell Dilution (X)		
			Vi-Cell ID:	VCD (X.X)	x 10 ⁶ cells/mL	
☐ Yes	☐ No			Viability (X)	%	
Was any pump tubing shifted?	☐ Yes ☐ N/A		Osmometer ID	Osmolality (X)		
			Has Perfusion criteria been met? (VCD ≥ 1.8 x 10 ⁶ cells/mL AND 48 ± 6 hours post Medium Dilution) Continue to Section 18 for Perfusion Initiation.			☐ Yes ☐ No
Performed By/Date:			Witness By/Date:			

18 Seed Perfusion Initiation

Step	Description	Performed By/Date	Witnessed By/Date								
18.1	Ensure Steps 10.8 - 10.12 have been completed before proceeding.										
18.2	<u>Investigational Sample:</u> - Obtain a sampling assembly (6000391). - Record the assembly and welder information. - Label the assembly with "INT-034", 8012420 and batch number, bioreactor ID, "100 L Bioreactor Pre-Perfusion Initiation", initials, and date. - Weld the assembly to the sample port per SOP-0107187 or SOP-0107297. - Record retain sample per Attachment 7: Retain Samples. <table border="1" data-bbox="203 772 1156 1060"> <tr> <td>Welder ID</td><td></td></tr> <tr> <td>Part No.:</td><td></td></tr> <tr> <td>Batch No.:</td><td></td></tr> <tr> <td>Exp. Date:</td><td></td></tr> </table> SAP Consumption By/Date	Welder ID		Part No.:		Batch No.:		Exp. Date:			
Welder ID											
Part No.:											
Batch No.:											
Exp. Date:											
18.3	- Dispense approximately 50 mL to be stored as investigation sample. - Seal off assembly for removal and record sealer information. <table border="1" data-bbox="203 1270 1156 1333"> <tr> <td>Sealer ID</td><td></td></tr> </table>	Sealer ID									
Sealer ID											
18.4	Verify the process trend for BRXFEED-4003 is activated.										

Step	Description								Performed By/Date	Witnessed By/Date	
18.5	Calculate the Perfusion Rate.										
	Total Cell Culture Volume (Step 16.14)	x	Perfusion Rate	x	1000 g/kg	÷	24 hrs/day	=	Perfusion Rate (X.XX)		
	kg		0.30 VVD						g/hr		
18.6	Input the following information into the feedcart per SOP-0126942.										
	Input N-1 Feedcart Settings										
	Feed cart ID		Step 12.2								
	N-1 Formula		IFBN1 PermG								
	N-1 Recipe		Feeds								
	Pump State		Inactive								
	EM Module		None								
	Recipe Inputs										
	TCCV Working Volume		Step 16.14								
	Perfusion Rate		Step 18.5								
	Feed Cart Pump Settings		Gravimetric								
	Input Alarms										
	Alarm Type			Limits			Delay (ms)				
	Pump Speed Alarm		HH	210 RPM			30,000				
	Gravimetric Alarm		DevA	50 g/hr			300,000				
	Input N-1 Recipe Density Check										
	Perfusion Feeds				Density (kg/L)						
	Permeate				1						

Step	Description			Performed By/Date	Witnessed By/Date																																				
18.7	Ensure the following Perfusion Start Criteria has been met.																																								
	VCD ($\geq 1.8 \times 10^6$ cells/mL)	_____ x 10 ⁶ cells/mL																																							
	Hours Post Dilution (Step 16.12)	_____ (48 ± 6 hours)																																							
18.8	- Unclamp the Permeate Line to the Permeate Waste Vessel. - Ensure the TFF 1 Permeate Line is open. - Ensure the TFF 1 Permeate Line is primed per SOP-0126942.																																								
18.9	Load the batch recipe “ FEEDS ” and Feed cart formula “ IFBN1 PermG ” per SOP-0126942.																																								
18.10	- Record the Perfusion Medium (PN: 8012342) pump below. - Prime the Perfusion Medium addition line per SOP-0126420. - Input the following level control volume in the ABEC.																																								
	Addition Pump	N-1 Level Control Max Weight CP	Select Status																																						
	_____	Bioreactor Weight After Dilution (Step 16.11)	Level Control																																						
18.11	Ensure the following Levitronix 1 alarms are enabled.																																								
	<table><tr><th colspan="4">Levitronix 1 Alarms</th></tr><tr><th></th><th>Low</th><th>High</th><th>Alarm Status</th></tr><tr><td>Pump Speed (RPM)</td><td>600</td><td>3000</td><td>Enable</td></tr><tr><td>Flow rate (l/hr)</td><td>2.4</td><td>6.4</td><td>Enable</td></tr><tr><td>Temp (°C)</td><td></td><td></td><td>Disable</td></tr><tr><td>TMP (psi)</td><td>0</td><td>0.5</td><td>Enabled</td></tr><tr><td>P1 (psi)</td><td>0</td><td>3</td><td>Enable</td></tr><tr><td>P2 (psi)</td><td>0</td><td>3</td><td>Enable</td></tr><tr><td>P3 (psi)</td><td>0</td><td>3</td><td>Enable</td></tr></table>					Levitronix 1 Alarms					Low	High	Alarm Status	Pump Speed (RPM)	600	3000	Enable	Flow rate (l/hr)	2.4	6.4	Enable	Temp (°C)			Disable	TMP (psi)	0	0.5	Enabled	P1 (psi)	0	3	Enable	P2 (psi)	0	3	Enable	P3 (psi)	0	3	Enable
	Levitronix 1 Alarms																																								
		Low	High			Alarm Status																																			
	Pump Speed (RPM)	600	3000			Enable																																			
	Flow rate (l/hr)	2.4	6.4			Enable																																			
	Temp (°C)					Disable																																			
	TMP (psi)	0	0.5			Enabled																																			
	P1 (psi)	0	3			Enable																																			
	P2 (psi)	0	3			Enable																																			
	P3 (psi)	0	3			Enable																																			
18.12	Record time and date of Perfusion Initiation.																																								
	Time/Date of Perfusion Initiation																																								
18.13	N-1 Perfusion Phase Sampling will begin day after Perfusion Initiation.																																								

19 Daily Sampling – Perfusion Phase

Step	Description	Performed By/Date	Witnessed By/Date
19.1	On a daily basis, including Day 0 or as instructed by supervisor/designee: • Prepare sample assembly by flushing 50 mL per SOP-0126420 • Collect approximately 10 mL sample from the bioreactor per SOP-0126420 and perform the following analysis: – Perform offline pH check per SOP-0107252. – Perform a one point correction of pH probes per SOP-0107196 when required. NOTE: If one point correction is required, it must be performed prior to any additions. – Perform offline osmolality check per SOP-0080509. – Measure cell density and percent viability per SOP-0115565. NOTE: Use CHO Method 2 cell type for cell count with Vi-CELL blu. • Perform glucose and lactate analysis • Record bioreactor conditions in Table 3: Perfusion Bioreactor Conditions N-1. • If the permeate bag needs to be replaced utilize Attachment 2: Permeate Bag Replacement to document the actions taken. • If needed, additional sample pages can be found in Attachment 9: Additional Sampling Pages (N-1).		

Table 3: Perfusion Bioreactor Conditions N-1

Perform daily sampling of the bioreactor.			BGA ID:		Measurements		Result	
Day	Seed Day _____				pH (T) (X.XX)			
Time					pCO ₂ (T) (X.X)		mmHg	
Choose the primary DO probe	☐ AIT-01 / ☐ AIT-03				pO ₂ (T) (X.X)		mmHg	
Bioreactor Temp. (°C) (XX.X) (35.0 - 38.0°C)					Bioreactor Glu (X)		mg/dL	
Pressure (psig) (X.XX) (0 - 0.6 psig)			Lac (mmol/L) (X.X)	Lactate Dilution	Calculated Lactate (mmol/L) (X.X)			
pO ₂ (%) (X.X) (10 - 110%)				☐ N/A	☐ N/A			
Bioreactor pH (X.XX) (6.70 - 7.50)			Adjustment <i>Perform a one-point calibration if pH difference is < -0.05 or > 0.05. Take a 2nd pH sample if pH difference is < -0.10 or > 0.10. Record information from both probes. Only primary probe results will be used for processing decisions. The secondary probe results are for information only (FIO).</i>					
Agitation (rpm) (X) (60 - 150 rpm)			Select Primary Probe	Bioreactor pH (X.XX)	Offline pH (T) (X.XX)	pH Difference		
Overlay Air (slpm) (X.X) (1 - 10 slpm)			AIT-02		-	=		
			AIT-04					
Air Sparge (slpm) (X.X) (0.5 - 5.0 slpm)			Is one-point calibration required based on 1 st pH sample results?				☐ Yes ☐ No	
O ₂ Sparge (slpm) (X.X)			Is a 2 nd pH sample required? If yes, go to Attachment 6: Additional pH Monitoring – pH samples' information.				☐ Yes ☐ No	
CO ₂ Sparge (slpm) (X.X)			Pipettor ID (if applicable)	Measurements	Results			
Bioreactor Weight (kg) (X)				Vi-Cell Dilution (X)				
			Vi-Cell ID:	VCD (X.X)	x 10 ⁶ cells/mL			
Foam Check: Is Antifoam addition Needed? Proceed to Attachment 4 for manual additions.		☐ No ☐ Yes		Viability (X)	%			
P1	P2	P3	Osmometer ID	Osmolality (X)				
psi	psi	psi						
TMP (≤ 1.0 psi)		psi		Is Perfusion Rate Change Required? (72 ± 6 hours post Medium Dilution) Continue to Section 20 .				☐ Yes ☐ No
Was any pump tubing shifted?		☐ Yes ☐ N/A						
Performed By/Date:					Witness By/Date:			

Table 3: Perfusion Bioreactor Conditions N-1

Perform daily sampling of the bioreactor.			BGA ID:		Measurements		Result	
Day	Seed Day _____				pH (T) (X.XX)			
Time					pCO ₂ (T) (X.X)		mmHg	
Choose the primary DO probe	☐ AIT-01 / ☐ AIT-03				pO ₂ (T) (X.X)		mmHg	
Bioreactor Temp. (°C) (XX.X) (35.0 - 38.0°C)					Bioreactor Glu (X)		mg/dL	
Pressure (psig) (X.XX) (0 - 0.6 psig)			Lac (mmol/L) (X.X)	Lactate Dilution	Calculated Lactate (mmol/L) (X.X)			
pO ₂ (%) (X.X) (10 - 110%)				☐ N/A	☐ N/A			
Bioreactor pH (X.XX) (6.70 - 7.50)			Adjustment <i>Perform a one-point calibration if pH difference is < -0.05 or > 0.05. Take a 2nd pH sample if pH difference is < -0.10 or > 0.10. Record information from both probes. Only primary probe results will be used for processing decisions. The secondary probe results are for information only (FIO).</i>					
Agitation (rpm) (X) (60 - 150 rpm)			Select Primary Probe	Bioreactor pH (X.XX)	Offline pH (T) (X.XX)	pH Difference		
Overlay Air (slpm) (X.X) (1 - 10 slpm)			AIT-02		-	=		
			AIT-04					
Air Sparge (slpm) (X.X) (0.5 - 5.0 slpm)			Is one-point calibration required based on 1 st pH sample results?				☐ Yes ☐ No	
O ₂ Sparge (slpm) (X.X)			Is a 2 nd pH sample required? If yes, go to Attachment 6: Additional pH Monitoring – pH samples' information.				☐ Yes ☐ No	
CO ₂ Sparge (slpm) (X.X)			Pipettor ID (if applicable)	Measurements	Results			
Bioreactor Weight (kg) (X)				Vi-Cell Dilution (X)				
			Vi-Cell ID:	VCD (X.X)	x 10 ⁶ cells/mL			
Foam Check: Is Antifoam addition Needed? Proceed to Attachment 4 for manual additions.		☐ No ☐ Yes		Viability (X)	%			
P1	P2	P3	Osmometer ID	Osmolality (X)				
psi	psi	psi						
TMP (≤ 1.0 psi)		psi		Is Perfusion Rate Change Required? (96 ± 6 hours post Medium Dilution) Continue to Section 20 .			☐ Yes ☐ No	
Was any pump tubing shifted?		☐ Yes ☐ N/A						
Performed By/Date:			Witness By/Date:					

Table 3: Perfusion Bioreactor Conditions N-1

Perform daily sampling of the bioreactor.			BGA ID:	Measurements	Result	
Day	Seed Day _____			pH (T) (X.XX)		
Time				pCO ₂ (T) (X.X)	mmHg	
Choose the primary DO probe	☐ AIT-01 / ☐ AIT-03			pO ₂ (T) (X.X)	mmHg	
Bioreactor Temp. (°C) (XX.X) (35.0 - 38.0°C)				Bioreactor Glu (X)	mg/dL	
Pressure (psig) (X.XX) (0 - 0.6 psig)			Lac (mmol/L) (X.X)	Lactate Dilution	Calculated Lactate (mmol/L) (X.X)	
pO ₂ (%) (X.X) (10 - 110%)				☐ N/A	☐ N/A	
Bioreactor pH (X.XX) (6.70 - 7.50)			Adjustment <i>Perform a one-point calibration if pH difference is < -0.05 or > 0.05. Take a 2nd pH sample if pH difference is < -0.10 or > 0.10. Record information from both probes. Only primary probe results will be used for processing decisions. The secondary probe results are for information only (FIO).</i>			
Agitation (rpm) (X) (60 - 150 rpm)			Select Primary Probe	Bioreactor pH (X.XX)	Offline pH (T) (X.XX)	pH Difference
Overlay Air (slpm) (X.X) (1 - 10 slpm)			AIT-02		-	=
			AIT-04			
Air Sparge (slpm) (X.X) (0.5 - 5.0 slpm)			Is one-point calibration required based on 1 st pH sample results?			☐ Yes ☐ No
O ₂ Sparge (slpm) (X.X)			Is a 2 nd pH sample required? If yes, go to Attachment 6: Additional pH Monitoring – pH samples' information.			☐ Yes ☐ No
CO ₂ Sparge (slpm) (X.X)			Pipettor ID (if applicable)	Measurements	Results	
Bioreactor Weight (kg) (X)				Vi-Cell Dilution (X)		
			Vi-Cell ID:	VCD (X.X)	x 10 ⁶ cells/mL	
Foam Check: Is Antifoam addition Needed? Proceed to Attachment 4 for manual additions.		☐ No ☐ Yes		Viability (X)		
P1	P2	P3	Osmometer ID	Osmolality (X)		
psi	psi	psi				
TMP (≤ 1.0 psi)	psi		Has Transfer Criteria been met? (VCD ≥ 26.0 x 10 ⁶ cells/mL) Continue to Section 21 .			☐ Yes ☐ No
			Is Perfusion Rate Change Required?*			☐ Yes ☐ No
Was any pump tubing shifted?		☐ Yes ☐ N/A	*If Transfer Criteria has been met, Perfusion Rate Change is not required. Proceed to Section 21 .			
Performed By/Date:				Witness By/Date:		

Table 3: Perfusion Bioreactor Conditions N-1

Perform daily sampling of the bioreactor.			BGA ID:		Measurements		Result	
Day	Seed Day _____				pH (T) (X.XX)			
Time					pCO ₂ (T) (X.X)		mmHg	
Choose the primary DO probe	☐ AIT-01 / ☐ AIT-03				pO ₂ (T) (X.X)		mmHg	
Bioreactor Temp. (°C) (XX.X) (35.0 - 38.0°C)					Bioreactor Glu (X)		mg/dL	
Pressure (psig) (X.XX) (0 - 0.6 psig)			Lac (mmol/L) (X.X)	Lactate Dilution	Calculated Lactate (mmol/L) (X.X)			
pO ₂ (%) (X.X) (10 - 110%)				☐ N/A	☐ N/A			
Bioreactor pH (X.XX) (6.70 - 7.50)			Adjustment Perform a one-point calibration if pH difference is < -0.05 or > 0.05. Take a 2nd pH sample if pH difference is < -0.10 or > 0.10. Record information from both probes. Only primary probe results will be used for processing decisions. The secondary probe results are for information only (FIO).					
Agitation (rpm) (X) (60 - 150 rpm)			Select Primary Probe	Bioreactor pH (X.XX)	Offline pH (T) (X.XX)	pH Difference		
Overlay Air (slpm) (X.X) (1 - 10 slpm)			AIT-02		-	=		
			AIT-04					
Air Sparge (slpm) (X.X) (0.5 - 5.0 slpm)			Is one-point calibration required based on 1 st pH sample results?				☐ Yes ☐ No	
O ₂ Sparge (slpm) (X.X)			Is a 2 nd pH sample required? If yes, go to Attachment 6: Additional pH Monitoring – pH samples' information.				☐ Yes ☐ No	
CO ₂ Sparge (slpm) (X.X)			Pipettor ID (if applicable)	Measurements	Results			
Bioreactor Weight (kg) (X)				Vi-Cell Dilution (X)				
			Vi-Cell ID:	VCD (X.X)	x 10 ⁶ cells/mL			
Foam Check: Is Antifoam addition Needed? Proceed to Attachment 4 for manual additions.		☐ No ☐ Yes		Viability (X)	%			
P1	P2	P3	Osmometer ID	Osmolality (X)				
psi	psi	psi						
TMP (≤ 1.0 psi)		psi		Has Transfer Criteria been met? (VCD ≥ 26.0 x 10 ⁶ cells/mL) Continue to Section 21.				☐ Yes ☐ No
Was any pump tubing shifted?		☐ Yes ☐ N/A						
Performed By/Date:			Witness By/Date:					

Table 3: Perfusion Bioreactor Conditions N-1

Perform daily sampling of the bioreactor.			BGA ID:		Measurements		Result	
Day	Seed Day _____				pH (T) (X.XX)			
Time					pCO ₂ (T) (X.X)		mmHg	
Choose the primary DO probe	☐ AIT-01 / ☐ AIT-03				pO ₂ (T) (X.X)		mmHg	
Bioreactor Temp. (°C) (XX.X) (35.0 - 38.0°C)					Bioreactor Glu (X)		mg/dL	
Pressure (psig) (X.XX) (0 – 0.6 psig)			Lac (mg/dL) (X.X)	Lactate Dilution	Calculated Lactate (mg/dL) (X.X)			
pO ₂ (%) (X.X) (10 - 110%)				☐ N/A	☐ N/A			
Bioreactor pH (X.XX) (6.70 - 7.50)			Adjustment <i>Perform a one-point calibration if pH difference is < -0.05 or > 0.05. Take a 2nd pH sample if pH difference is < -0.10 or > 0.10. Record information from both probes. Only primary probe results will be used for processing decisions. The secondary probe results are for information only (FIO).</i>					
Agitation (rpm) (X) (60 - 150 rpm)			Select Primary Probe	Bioreactor pH (X.XX)	Offline pH (T) (X.XX)	pH Difference		
Overlay Air (slpm) (X.X) (1 - 10 slpm)			AIT-02		-	=		
			AIT-04					
Air Sparge (slpm) (X.X) (0.5 - 5.0 slpm)			Is one-point calibration required based on 1 st pH sample results?				☐ Yes ☐ No	
O ₂ Sparge (slpm) (X.X)			Is a 2 nd pH sample required? If yes, go to Attachment 6: Additional pH Monitoring – pH samples' information.				☐ Yes ☐ No	
CO ₂ Sparge (slpm) (X.X)			Pipettor ID (if applicable)	Measurements	Results			
Bioreactor Weight (kg) (X)				Vi-Cell Dilution (X)				
			Vi-Cell ID:	VCD (X.X)	x 10 ⁶ cells/mL			
Foam Check: Is Antifoam addition Needed? Proceed to Attachment 4 for manual additions.		☐ No ☐ Yes		Viability (X)	%			
P1	P2	P3	Osmometer ID	Osmolality (X)				
psi	psi	psi						
TMP (≤ 1.0 psi)		psi		Has Transfer Criteria been met? (VCD ≥ 26.0 x 10 ⁶ cells/mL) Continue to Section 21.				☐ Yes ☐ No
Was any pump tubing shifted?		☐ Yes ☐ N/A						
Performed By/Date:					Witness By/Date:			

Table 3: Perfusion Bioreactor Conditions N-1

Perform daily sampling of the bioreactor.			BGA ID:	Measurements	Result	
Day	Seed Day _____			pH (T) (X.XX)		
Time				pCO ₂ (T) (X.X)	mmHg	
Choose the primary DO probe	☐ AIT-01 / ☐ AIT-03			pO ₂ (T) (X.X)	mmHg	
Bioreactor Temp. (°C) (XX.X) (35.0 - 38.0°C)				Bioreactor Glu (X)	mg/dL	
Pressure (psig) (X.XX) (0 – 0.6 psig)			Lac (mg/dL) (X.X)	Lactate Dilution	Calculated Lactate (mg/dL) (X.X)	
pO ₂ (%) (X.X) (10 - 110%)				☐ N/A	☐ N/A	
Bioreactor pH (X.XX) (6.70 - 7.50)			Adjustment <i>Perform a one-point calibration if pH difference is < -0.05 or > 0.05. Take a 2nd pH sample if pH difference is < -0.10 or > 0.10. Record information from both probes. Only primary probe results will be used for processing decisions. The secondary probe results are for information only (FIO).</i>			
Agitation (rpm) (X) (60 - 150 rpm)			Select Primary Probe	Bioreactor pH (X.XX)	Offline pH (T) (X.XX)	pH Difference
Overlay Air (slpm) (X.X) (1 - 10 slpm)			AIT-02		-	=
			AIT-04			
Air Sparge (slpm) (X.X) (0.5 - 5.0 slpm)			Is one-point calibration required based on 1 st pH sample results?			☐ Yes ☐ No
O ₂ Sparge (slpm) (X.X)			Is a 2 nd pH sample required? If yes, go to Attachment 6: Additional pH Monitoring – pH samples' information.			☐ Yes ☐ No
CO ₂ Sparge (slpm) (X.X)			Pipettor ID (if applicable)	Measurements	Results	
Bioreactor Weight (kg) (X)				Vi-Cell Dilution (X)		
			Vi-Cell ID:	VCD (X.X)	x 10 ⁶ cells/mL	
Foam Check: Is Antifoam addition Needed? Proceed to Attachment 4 for manual additions.		☐ No ☐ Yes		Viability (X)		
P1	P2	P3	Osmometer ID	Osmolality (X)		
psi	psi	psi				
TMP (≤ 1.0 psi)		psi		Has Transfer Criteria been met? (VCD ≥ 26.0 x 10 ⁶ cells/mL) Continue to Section 21.		
Was any pump tubing shifted?		☐ Yes ☐ N/A		☐ Yes ☐ No		
Performed By/Date:			Witness By/Date:			

20 Perfusion Rate Change

Step	Description								Performed By/Date	Witnessed By/Date	
20.1	- When the hours post Dilution (Step 16.12) is equal to 72 ± 6 hours, change the perfusion rate. - Calculate the new Perfusion Rate.										
	Total Cell Culture Volume (Step 16.14)	x	Perfusion Rate	x	1000 g/kg	÷	24 hrs/day	=			Perfusion Rate (X.XX)
	kg		0.70 VVD								g/hr
20.2	Input the Perfusion Rate into the feedcart per SOP-0126942.										
	Hours Since Dilution End				_____ (72 ± 6 hours)						
	Perfusion Rate				_____ (Step 20.1)						
20.3	- When the hours post Dilution (Step 16.12) is equal to 96 ± 6 hours, change the perfusion rate. - Calculate the new Perfusion Rate.										
	Total Cell Culture Volume (Step 16.14)	x	Perfusion Rate	x	1000 g/kg	÷	24 hrs/day	=			Perfusion Rate (X.XX)
	kg		1.50 VVD								g/hr
20.4	Input the Perfusion Rate into the feedcart per SOP-0126942.										
	Hours Since Dilution End				_____ (96 ± 6 hours)						
	Perfusion Rate				_____ (Step 20.3)						

Step	Description								Performed By/Date	Witnessed By/Date	
20.5	- When the hours post Dilution (Step 16.12) is equal to 120 ± 6 hours, change the perfusion rate. - Calculate the new Perfusion Rate.										
	Total Cell Culture Volume (Step 16.14)	x	Perfusion Rate	x	1000 g/kg	÷	24 hrs/day	=			Perfusion Rate (X.XX)
	kg		2.00 VVD								g/hr
20.6	Input the Perfusion Rate into the feedcart per SOP-0126942.										
	Hours Since Dilution End			_____ (120 ± 6 hours)							
	Perfusion Rate			_____ (Step 20.5)							

21 Transfer to 500 L STR

Step	Description	Performed By/Date	Witnessed By/Date		
21.1	When the viable cell density is $\geq 26.0 \times 10^6$ cells/mL, prepare to transfer to 500 L Production STR. NOTE: Use the CHO Method 2 for cell count with Vi-CELL BLU.				
	Seed Day:			Viable Cell Density (VCD):	Viability:
				_____ $\times 10^6$ cells/mL ($\geq 26.0 \times 10^6$ cells/mL)	_____ % (90%)
	Contact Supervisor if: - the viability is less than 90% and proceed with comment. - VCD is not met by Perfusion Day 5: - Additional samples can be taken under supervisor direction. Document daily sample results in Attachment 9: Additional Sampling Pages (N-1). NOTE: Maximum time allowed in Perfusion step is 7 days. - VCD is greater than 33.3×10^6 cells/mL.				
	Supervisor/designee contacted: _____				
21.2	<u>Pre-Transfer Investigational Sample:</u>				
	- Obtain a sampling assembly (6000391). - Record the assembly and welder information. - Label the assembly with "INT-034", 8012420 and batch number, bioreactor ID, "100 L Bioreactor Pre - 500L Transfer", initials, and date. - Weld the assembly to the sample port per SOP-0080596.				
	Welder ID				
	Part No.:				
	Batch No.:				
	Exp. Date:				
	SAP Consumption By/date				

Step	Description	Performed By/Date	Witnessed By/Date							
21.3	- Dispense approximately 50 mL to be stored as investigation sample. - Record the sealer information below. - Record retain sample per Attachment 7: Retain Samples.									
	<table border="1"> <tr> <td>Sealer ID</td><td></td></tr> </table>			Sealer ID						
Sealer ID										
21.4	- End the N-1 Feedcart Recipe batch per SOP-0126942. - Record the Time the recipe was stopped and permeate waste bag weight.									
	<table border="1"> <tr> <td>Time the Feedcart Recipe was Stopped</td><td>Gross Weight of Permeate Waste</td><td>Net Weight of Permeate Waste*</td></tr> <tr> <td></td><td>kg</td><td>kg</td></tr> </table>			Time the Feedcart Recipe was Stopped	Gross Weight of Permeate Waste	Net Weight of Permeate Waste*		kg	kg	
	Time the Feedcart Recipe was Stopped			Gross Weight of Permeate Waste	Net Weight of Permeate Waste*					
				kg	kg					
*Net Weight of Permeate Waste = Gross Weight of Permeate Waste (above) – Tare Weight of Permeate Vessel (Step 10.4 or Attachment 2) NOTE: Use Tare Weight of Permeate Vessel from Attachment 2 if permeate bag was replaced during run.										
21.5	- Stop the Perfusion Medium pump per SOP-0126420. - Close the permeate line. - Remove the Perfusion Medium from the ABEC per SOP-0080522.									
	<table border="1"> <tr> <td>Sealer ID</td><td></td></tr> </table>			Sealer ID						
Sealer ID										
21.6	- Obtain the gross weight of the Perfusion Medium after use. - Calculate the net weight of PN: 8012342 used during the run.									
	<table border="1"> <tr> <td>Scale ID</td><td></td></tr> </table>			Scale ID						
	Scale ID									
	<table border="1"> <tr> <td>Calibration Due Date</td><td></td></tr> </table>			Calibration Due Date						
	Calibration Due Date									
	<table border="1"> <tr> <td>Gross Weight of CCM-FR-2 Perfusion Medium (M-051-V4) (PN: 8012342) after use</td><td></td></tr> </table>			Gross Weight of CCM-FR-2 Perfusion Medium (M-051-V4) (PN: 8012342) after use						
	Gross Weight of CCM-FR-2 Perfusion Medium (M-051-V4) (PN: 8012342) after use									
<table border="1"> <tr> <td>Net Weight of CCM-FR-2 Perfusion Medium (M-051-V4) (PN: 8012342) used*</td><td></td></tr> </table>	Net Weight of CCM-FR-2 Perfusion Medium (M-051-V4) (PN: 8012342) used*									
Net Weight of CCM-FR-2 Perfusion Medium (M-051-V4) (PN: 8012342) used*										
*Net Weight of 8012342 used = Gross Weight of 8012342 before use (Step 10.9) – Gross Weight of 8012342 after use (above)										
<table border="1"> <tr> <td>SAP Consumption By/Date</td><td></td></tr> </table>		SAP Consumption By/Date								
SAP Consumption By/Date										

Step	Description	Performed By/Date	Witnessed By/Date														
21.7	- Record the weight of the antifoam bag. - Calculate the total amount of antifoam added during the run. NOTE: Ensure the antifoam bag tubing is in the same configuration as when the initial weight was taken in Step 10.7. <table border="1"> <tr> <td>Antifoam Bag Weight After Priming</td><td>-</td><td>Antifoam Bag Final Weight</td><td>=</td><td>Weight of Antifoam Added (X)</td></tr> <tr> <td>g</td><td></td><td>g</td><td></td><td>g</td></tr> </table> NOTE: The maximum antifoam amount allowed is 330 g for PN: 4102603/4102604 or 165 g for PN: 4102057.	Antifoam Bag Weight After Priming	-	Antifoam Bag Final Weight	=	Weight of Antifoam Added (X)	g		g		g						
Antifoam Bag Weight After Priming	-	Antifoam Bag Final Weight	=	Weight of Antifoam Added (X)													
g		g		g													
21.8	165 g for PN: 4102057 Is the Weight of Antifoam Added (Step) > OR 330 g for PN: 4102063 or 4102604? <table border="1"> <tr> <td>☐ Yes</td><td>Proceed to the next step</td></tr> <tr> <td>☐ No</td><td>N/A Steps 21.9 through 21.12 and continue to Step 21.13.</td></tr> </table>	☐ Yes	Proceed to the next step	☐ No	N/A Steps 21.9 through 21.12 and continue to Step 21.13 .												
☐ Yes	Proceed to the next step																
☐ No	N/A Steps 21.9 through 21.12 and continue to Step 21.13 .																
21.9	Record the concentration of simethicone in the antifoam from the manufacturer's Certificate of Analysis below: <table border="1"> <tr> <td>Concentration of Simethicone</td><td>_____ %</td></tr> </table>	Concentration of Simethicone	_____ %														
Concentration of Simethicone	_____ %																
21.10	Calculate the total amount of simethicone added. <table border="1"> <tr> <td>Weight of Antifoam Added (Step 21.7)</td><td>x</td><td>Concentration of Simethicone (Step 21.9)</td><td>÷</td><td>100</td><td>=</td><td>Total Amount of Simethicone Added (X)</td></tr> <tr> <td>g</td><td></td><td>%</td><td></td><td></td><td></td><td>g</td></tr> </table>	Weight of Antifoam Added (Step 21.7)	x	Concentration of Simethicone (Step 21.9)	÷	100	=	Total Amount of Simethicone Added (X)	g		%				g		
Weight of Antifoam Added (Step 21.7)	x	Concentration of Simethicone (Step 21.9)	÷	100	=	Total Amount of Simethicone Added (X)											
g		%				g											
21.11	Calculate the concentration of simethicone in the pre-harvest product. <table border="1"> <tr> <td>Total Amount of Simethicone Added (Step 21.10)</td><td>÷</td><td>TCCV (Step 16.14)</td><td>x</td><td>1,000</td><td>=</td><td>Concentration of Simethicone at the end of N-1 (X)</td></tr> <tr> <td>g</td><td></td><td>kg</td><td></td><td></td><td></td><td>ppm</td></tr> </table>	Total Amount of Simethicone Added (Step 21.10)	÷	TCCV (Step 16.14)	x	1,000	=	Concentration of Simethicone at the end of N-1 (X)	g		kg				ppm		
Total Amount of Simethicone Added (Step 21.10)	÷	TCCV (Step 16.14)	x	1,000	=	Concentration of Simethicone at the end of N-1 (X)											
g		kg				ppm											
21.12	If concentration of simethicone at the end of N-1 in Step 21.11 exceeds 50 ppm, contact Area Manager and QA. Record decision(s) in Attachment 10: Comments Section. Area Manager/Designee contacted: _____																

Step	Description	Performed By/Date	Witnessed By/Date			
21.13	Prior to transfer, disable the alarms below.					
	Parameter	Low		High	Deviation	Alarm Status
	Vessel Temp (°C)	35.0		38.0	1.0	Disable
	Vessel Weight (kg)	85		50	115	Disable
	Agitation (rpm)	60		150		Disable
	pH	6.70		7.50	0.22	Disable
	DO (%)	10		110	10	Disable
	Levitronix 1 Alarms					
	Pump Speed (RPM)	600		3000		Disable
	Flow rate (l/hr)	2.4		6.4		Disable
	Temp (°C)					Disable
	TMP (psi)	0		0.5		Disable
	P1 (psi)	0		3		Disable
	P2 (psi)	0		3		Disable
	P3 (psi)	0		3		Disable
	21.14	On the Recipe Alarms page, change the Interlock alarm per the table below:				
Bioreactor System Alarm		Low Interlock				
Vessel Weight (kg)		0				

Step	Description	Performed By/Date	Witnessed By/Date																																							
21.15	On the Loops screen, set the parameters below prior to transfer.																																									
	<table><tr><th>Loop</th><th>Setpoint</th><th>Status</th></tr><tr><td>Vessel Temp SP</td><td>N/A</td><td>OFF</td></tr><tr><td>pH SP</td><td>N/A</td><td>OFF</td></tr><tr><td>pH deadband</td><td>N/A</td><td>OFF</td></tr><tr><td>DO SP</td><td>50%</td><td>Cascade</td></tr><tr><td>Agitation SP</td><td>75 rpm</td><td>Cascade</td></tr><tr><td>Air Overlay SP</td><td>5.0 slpm</td><td>AUTO</td></tr><tr><td>Primary Air Sparge SP</td><td>0.5 slpm</td><td>AUTO</td></tr><tr><td>Secondary Air Sparge SP</td><td>0</td><td>OFF</td></tr><tr><td>Primary O2 Sparge SP</td><td>0</td><td>OFF</td></tr><tr><td>Secondary O2 Sparge SP</td><td>0</td><td>OFF</td></tr><tr><td>pH Strategy</td><td></td><td>OFF</td></tr><tr><td>DO Strategy</td><td></td><td>Profile</td></tr></table>			Loop	Setpoint	Status	Vessel Temp SP	N/A	OFF	pH SP	N/A	OFF	pH deadband	N/A	OFF	DO SP	50%	Cascade	Agitation SP	75 rpm	Cascade	Air Overlay SP	5.0 slpm	AUTO	Primary Air Sparge SP	0.5 slpm	AUTO	Secondary Air Sparge SP	0	OFF	Primary O2 Sparge SP	0	OFF	Secondary O2 Sparge SP	0	OFF	pH Strategy		OFF	DO Strategy		Profile
	Loop			Setpoint	Status																																					
	Vessel Temp SP			N/A	OFF																																					
	pH SP			N/A	OFF																																					
	pH deadband			N/A	OFF																																					
	DO SP			50%	Cascade																																					
	Agitation SP			75 rpm	Cascade																																					
	Air Overlay SP			5.0 slpm	AUTO																																					
	Primary Air Sparge SP			0.5 slpm	AUTO																																					
	Secondary Air Sparge SP			0	OFF																																					
	Primary O2 Sparge SP			0	OFF																																					
	Secondary O2 Sparge SP			0	OFF																																					
	pH Strategy				OFF																																					
DO Strategy		Profile																																								
21.16	Proceed to PN 8012421 / INT-034 - 500 L Production Bioreactor (GPF-North, 500 L, Process 1)																																									
21.17	- Monitor the liquid level in the bioreactor throughout transfer to the 500 L STR. - End Grow 2 Recipe once liquid level reaches the probe belt.																																									
21.18	Stop TFF recirculation once liquid level reaches the DC supply line and flow from return line has stopped.																																									

22 Post-Process Procedures

Step	Description	Performed by/Date	Witnessed by/Date
22.1	Ensure the TFF is stopped and broken down per SOP-0126879 and SOP-0126420.		
22.2	Ensure bioreactor breakdown is performed per SOP-0126420.		

23 Batch Record Review

Step	Description	Initials/Date
23.1	Verify that all the following data is attached to this batch record: • Sample Assembly Attachment and Sampling (Attachment 1) • Permeate Bag Replacement (Attachment 2) • Additional Exhaust Filter Assembly (Attachment 3) • Antifoam Addition (Attachment 4) • Non-Routine Sampling Record (Attachment 5) • Additional pH Monitoring – pH Samples' Information (Attachment 6) • Retain Samples (Attachment 7) • Additional Sampling Pages (N-2) (Attachment 8) • Additional Sampling Pages (N-1) (Attachment 9) • Comments Section (Attachment 10)	
23.2	Verify that the Seed Bioreactor product net weight (Step 16.14) is correctly input into SAP.	
23.3	Review the Master Production Record for completeness, and accuracy sign the front page, and submit for review.	

Attachment 1: Sample Assembly Attachment and Sampling

Sampling Device				
Part No.:				
Batch No.:				
Expiration Date:				
Welder ID:				
Performed By/Date:				
Witnessed By/Date:				
Sampling Information				
Working Day or Process:				
Time:				
Approximate Volume:				
Sealer ID:				
Performed By/Date:				
Witnessed By/Date:				
SAP Consumption Performed by/Date:				

Sampling Device				
Part No.:				
Batch No.:				
Expiration Date:				
Welder ID:				
Performed By/Date:				
Witnessed By/Date:				
Sampling Information				
Working Day or Process:				
Time:				
Approximate Volume:				
Sealer ID:				
Performed By/Date:				
Witnessed By/Date:				
SAP Consumption Performed by/Date:				

Attachment 1: Sample Assembly Attachment and Sampling (continued)

Sampling Device				
Part No.:				
Batch No.:				
Expiration Date:				
Welder ID:				
Performed By/Date:				
Witnessed By/Date:				
Sampling Information				
Working Day or Process:				
Time:				
Approximate Volume:				
Sealer ID:				
Performed By/Date:				
Witnessed By/Date:				
SAP Consumption Performed by/Date:				

Sampling Device				
Part No.:				
Batch No.:				
Expiration Date:				
Welder ID:				
Performed By/Date:				
Witnessed By/Date:				
Sampling Information				
Working Day or Process:				
Time:				
Approximate Volume:				
Sealer ID:				
Performed By/Date:				
Witnessed By/Date:				
SAP Consumption Performed by/Date:				

Attachment 2: Permeate Bag Replacement

Permeate Bag Day of Replacement Information				
Working Day or Process:				
Time:				
Gross Weight of Permeate Waste Vessel before removal:				
Net Weight of Permeate Waste Vessel before removal*:				
Sealer ID:				
Performed By/Date:				
Witnessed By/Date:				
Permeate Bag Information				
Part No.:				
Batch No.:				
Expiration Date:				
Welder ID:				
Tare Weight of Permeate Vessel after attaching to DC**:				
Performed By/Date:				
Witnessed By/Date:				
SAP Consumption Performed by/Date:				

* Net Weight of Permeate Waste Vessel before removal = Gross Weight of Permeate Waste Vessel before removal – Tare Weight of Permeate Vessel after attaching to DC (**Step 10.4** or **Attachment 2**).

Ensure the same permeate scale from **Step 12.1 is used for the new permeate vessel.

Attachment 3: Additional Exhaust Filter Assembly

Exhaust Filter Assembly Information				
Exhaust Filter Assembly Part No.:				
Exhaust Filter Assembly Batch No.:				
Exhaust Filter Assembly Expiration Date:				
Tubing Assembly Part No.* :				
Tubing Assembly Batch No. * :				
Tubing Assembly Expiration Date * :				
Welder ID:				
Performed By/Date:				
Witnessed By/Date:				
SAP Consumption Performed by:				
Date Filter Clamp is opened:				
Performed By/Date:				
Witnessed By/Date:				

* N/A if additional tubing assembly was not used to connect Exhaust Filter to the DC.

Attachment 4: Antifoam Addition

Table 1: Manual Additions					
N-2 Phase					
Time of Addition	Balance ID ^(a)	Required Antifoam Amount (g)	Amount Added (g) (X):	Performed By/Date	Witnessed By/Date
	Calibration Due Date:	☐ 25 g ± 2 (4102057) ☐ 75 g ± 4 (4102603 or 4102604)			
	Calibration Due Date:				
	Calibration Due Date:				
	Calibration Due Date:				
N-1 Phase Pre-Perfusion Initiation					
Time of Addition	Balance ID ^(a)	Required Antifoam Amount (g)	Amount Added (g) (X):	Performed By/Date	Witnessed By/Date
	Calibration Due Date:	☐ 25 g ± 2 (4102057) ☐ 75 g ± 4 (4102603 or 4102604)			
	Calibration Due Date:				
	Calibration Due Date:				
	Calibration Due Date:				

(a) Ensure a balance weight check was performed within the last 7 days per SOP-0080224 or SOP-0080249.

Attachment 4: Antifoam Addition (continued)

Table 1: Manual Additions					
N-1 Phase Perfusion					
Time of Addition	Balance ID ^(a)	Required Antifoam Amount (g)	Amount Added (g) (X):	Performed By/Date	Witnessed By/Date
	Calibration Due Date:	☐ 25 g ± 2 (4102057) ☐ 75 g ± 4 (4102603 or 4102604)			
	Calibration Due Date:				
	Calibration Due Date:				
	Calibration Due Date:				
	Calibration Due Date:				
	Calibration Due Date:				
	Calibration Due Date:				
	Calibration Due Date:				
	Calibration Due Date:				
	Calibration Due Date:				
	Calibration Due Date:				
	Calibration Due Date:				
	Calibration Due Date:				
	Calibration Due Date:				

(a) Ensure a balance weight check was performed within the last 7 days per SOP-0080224 or SOP-0080249.

Attachment 5: Non-Routine Sampling Record

Process Section No.	Contents	Container Type	Quantity	Designation	Performed By/Date	Witnessed By/Date

Attachment 5: Non-Routine Sampling Record (continued)

Process Section No.	Contents	Container Type	Quantity	Designation	Performed By/Date	Witnessed By/Date

Attachment 6: Additional pH Monitoring - pH samples' information

Date	Sample Time	pH probe	Bioreactor pH (X.XX)	Bioreactor Temp. (°C) (XX.X)	BGA /pH Meter ID	Offline pH (X.XX)	pH Difference ($\frac{BRX\ pH - offline\ pH}{2}$)	Correction (Select type of correction)			Calibration New Value (X.XX)	Time performed	Performed By/Date	Witnessed By/Date
		<u>AIT-02</u>						☐ No	☐ full ⁽¹⁾	☐ ½ ⁽²⁾				
		<u>AIT-04</u>						☐ No	☐ full ⁽¹⁾	☐ ½ ⁽²⁾				
		<u>AIT-02</u>						☐ No	☐ full ⁽¹⁾	☐ ½ ⁽²⁾				
		<u>AIT-04</u>						☐ No	☐ full ⁽¹⁾	☐ ½ ⁽²⁾				
		<u>AIT-02</u>						☐ No	☐ full ⁽¹⁾	☐ ½ ⁽²⁾				
		<u>AIT-04</u>						☐ No	☐ full ⁽¹⁾	☐ ½ ⁽²⁾				
		<u>AIT-02</u>						☐ No	☐ full ⁽¹⁾	☐ ½ ⁽²⁾				
		<u>AIT-04</u>						☐ No	☐ full ⁽¹⁾	☐ ½ ⁽²⁾				

Instructions: Probes are standardized as needed independently of each other.

If pH difference after the 2nd sample is:

(1) $> \pm 0.05$ **and** $\leq \pm 0.10$, perform a full correction.

(2) **Still** $> \pm 0.10$, perform a ½ correction on the bioreactor transmitter per footnote (a) **and** notify supervisor/designee.

(a) Calculate ½ Correction value using the formula: $\frac{1}{2} \text{ Correction value} = (\text{Offline pH} + \text{Online pH}) \div 2$

(b) If a ½ correction is made, complete any required additions to the bioreactor. Once all additions are completed, **collect a 3rd pH sample** between 30 minutes and 4 hours after the additions were completed. **If pH difference is**

- $< \pm 0.05$, no correction is required.
- $\geq \pm 0.05$ and $\leq \pm 0.10$, correct bioreactor pH to match the offline pH value.
- $> \pm 0.10$, contact supervisor and record action taken in **Attachment 16**.

Attachment 6: Additional pH Monitoring – pH samples' information (continued)

Date	Sample Time	pH probe	Bioreactor pH (X.XX)	Bioreactor Temp. (°C) (XX.X)	BGA /pH Meter ID	Offline pH (X.XX)	pH Difference (<i>BRX pH – offline pH</i>)	Correction (<i>Select type of correction</i>)			Calibration New Value (X.XX)	Time performed	Performed By/Date	Witnessed By/Date
		<u>AIT-02</u>						☐ No	☐ full ⁽¹⁾	☐ ½ ⁽²⁾				
		<u>AIT-04</u>						☐ No	☐ full ⁽¹⁾	☐ ½ ⁽²⁾				
		<u>AIT-02</u>						☐ No	☐ full ⁽¹⁾	☐ ½ ⁽²⁾				
		<u>AIT-04</u>						☐ No	☐ full ⁽¹⁾	☐ ½ ⁽²⁾				
		<u>AIT-02</u>						☐ No	☐ full ⁽¹⁾	☐ ½ ⁽²⁾				
		<u>AIT-04</u>						☐ No	☐ full ⁽¹⁾	☐ ½ ⁽²⁾				
		<u>AIT-02</u>						☐ No	☐ full ⁽¹⁾	☐ ½ ⁽²⁾				
		<u>AIT-04</u>						☐ No	☐ full ⁽¹⁾	☐ ½ ⁽²⁾				
		<u>AIT-02</u>						☐ No	☐ full ⁽¹⁾	☐ ½ ⁽²⁾				
		<u>AIT-04</u>						☐ No	☐ full ⁽¹⁾	☐ ½ ⁽²⁾				
		<u>AIT-02</u>						☐ No	☐ full ⁽¹⁾	☐ ½ ⁽²⁾				
		<u>AIT-04</u>						☐ No	☐ full ⁽¹⁾	☐ ½ ⁽²⁾				

Instructions: Probes are standardized as needed independently of each other.

If pH difference after the 2nd sample is:

(1) $> \pm 0.05$ and $\leq \pm 0.10$, perform a full correction.

(2) *Still* $> \pm 0.10$, perform a ½ correction on the bioreactor transmitter per footnote (a) **and** notify supervisor/designee.

(a) Calculate ½ Correction value using the formula: $\frac{1}{2} \text{ Correction value} = (\text{Offline pH} + \text{Online pH}) \div 2$

(b) If a ½ correction is made, complete any required additions to the bioreactor. Once all additions are completed, **collect a 3rd pH sample** between 30 minutes and 4 hours after the additions were completed. **If pH difference is**

- $< \pm 0.05$, no correction is required.
- $\geq \pm 0.05$ and $\leq \pm 0.10$, correct bioreactor pH to match the offline pH value.
- $> \pm 0.10$, contact supervisor and record action taken in **Attachment 16**.

Attachment 7: Retain Samples

Process Step	Solution	Volume	Storage Temp. °C	Description	Notes	Sample collected?	Performed by/Date	Witnessed by/Date
SETUP	Seed Medium before charge (Step 11.5)	50 mL	2-8	Bioburden	Sample from bag prior to adding to ABEC	☐ Yes		
	Perfusion media (Step 10.11)	50 mL	2-8	Bioburden	Sample from bag prior to priming line	☐ Yes		
N-2	Medium sample before inoculation (Step 13.8)	50 mL	2-8	Bioburden	Sample from the ABEC prior to inoculation	☐ Yes		
N-1	Top off media before Dilution (Step 16.6)	50 mL	2-8	Bioburden	Sample from bag prior to adding to ABEC	☐ Yes		
	Culture before Dilution (Step 16.2)	50 mL	2-8	Cell containing Bioburden	Sample from the ABEC prior to dilution	☐ Yes		
	Culture before N-1 perfusion initiation (Step 18.2)	50 mL	2-8	Cell containing Bioburden	Sample from the ABEC prior to initiating N-1 perfusion	☐ Yes		
	Culture before transfer to 500 L STR (Step 21.2)	50 mL	2-8	Cell containing Bioburden	Sample from the ABEC prior to transferring to 500 L STR	☐ Yes		

Attachment 8: Additional Sampling Pages (N-2)

Perform daily sampling of the bioreactor.			BGA ID:	Measurements	Result	
Day	Seed Day _____			pH (T) (X.XX)		
Time				pCO ₂ (T) (X.X)	mmHg	
Select Primary DO Probe	☐ AIT-01 / ☐ AIT-03			pO ₂ (T) (X.X)	mmHg	
Bioreactor Temp. (°C) (XX.X) (35.0 - 38.0°C)				Bioreactor Glu (X)	mg/dL	
Pressure (psig) (X.XX) (0 – 0.6 psig)						
pO ₂ (%) (X.X) (10.0 - 110.0%)			Lac (mmol/L) (X.X)	Lactate Dilution	Calculated Lactate (mmol/L) (X.X)	
Bioreactor pH (X.XX) (6.85 - 7.50)				☐ N/A	☐ N/A	
Agitation (rpm) (X) (60 – 150 rpm)			Adjustment Perform a one-point calibration if pH difference is < -0.05 or > 0.05. Take a 2nd pH sample if pH difference is < -0.10 or > 0.10. Record information from both probes. Only primary probe results will be used for processing decisions. The secondary probe results are for information only (FIO).			
Overlay Air (slpm) (X.X) (1.0 – 10.0 slpm)			Select Primary pH Probe	Bioreactor pH (X.XX)	Offline pH (T) (X.XX)	pH Difference
Air Sparge (slpm) (X.X) (0.2 - 0.8 slpm)			☐ AIT-02		-	=
			☐ AIT-04			
O ₂ Sparge (slpm) (X.X)			Is one-point calibration required based on 1 st pH sample results?			☐ Yes ☐ No
CO ₂ Sparge (slpm) (X.X)			Is a 2 nd pH sample required? If yes, go to Attachment 6: Additional pH Monitoring – pH samples' information.			☐ Yes ☐ No
Bioreactor Weight (kg) (X)			Pipettor ID (if applicable)	Measurements	Results	
Foam Check: Is Antifoam addition Needed? Proceed to Attachment 4 for manual addition.				Vi-Cell Dilution (X)		
			Vi-Cell ID:	VCD (X.X)	x 10 ⁶ cells/mL	
☐ Yes	☐ No			Viability (X)	%	
Was any pump tubing shifted?	☐ Yes ☐ N/A		Osmometer ID	Osmolality (X)		
			Has Dilution criteria been met? (VCD ≥ 2.1 x 10 ⁶ cells/mL) Continue to Section 16 for Medium Dilution and N/A all subsequent sample pages			☐ Yes ☐ No
Performed By/Date:			Witness By/Date:			

Attachment 8: Additional Sampling Pages (N-2)

Perform daily sampling of the bioreactor.			BGA ID:	Measurements	Result	
Day	Seed Day _____			pH (T) (X.XX)		
Time				pCO ₂ (T) (X.X)	mmHg	
Select Primary DO Probe	☐ AIT-01 / ☐ AIT-03			pO ₂ (T) (X.X)	mmHg	
Bioreactor Temp. (°C) (XX.X) (35.0 - 38.0°C)				Bioreactor Glu (X)	mg/dL	
Pressure (psig) (X.XX) (0 - 0.6 psig)						
pO ₂ (%) (X.X) (10.0 - 110.0%)			Lac (mmol/L) (X.X)	Lactate Dilution	Calculated Lactate (mmol/L) (X.X)	
Bioreactor pH (X.XX) (6.85 - 7.50)				☐ N/A	☐ N/A	
Agitation (rpm) (X) (60 - 150 rpm)			Adjustment Perform a one-point calibration if pH difference is < -0.05 or > 0.05. Take a 2nd pH sample if pH difference is < -0.10 or > 0.10. Record information from both probes. Only primary probe results will be used for processing decisions. The secondary probe results are for information only (FIO).			
Overlay Air (slpm) (X.X) (1.0 - 10.0 slpm)			Select Primary pH Probe	Bioreactor pH (X.XX)	Offline pH (T) (X.XX)	pH Difference
Air Sparge (slpm) (X.X) (0.2 - 0.8 slpm)			☐ AIT-02		-	=
			☐ AIT-04			
O ₂ Sparge (slpm) (X.X)			Is one-point calibration required based on 1 st pH sample results?			☐ Yes ☐ No
CO ₂ Sparge (slpm) (X.X)			Is a 2 nd pH sample required? If yes, go to Attachment 6: Additional pH Monitoring – pH samples' information.			☐ Yes ☐ No
Bioreactor Weight (kg) (X)			Pipettor ID (if applicable)	Measurements	Results	
Foam Check: Is Antifoam addition Needed? Proceed to Attachment 4 for manual addition.				Vi-Cell Dilution (X)		
			Vi-Cell ID:	VCD (X.X)	x 10 ⁶ cells/mL	
☐ Yes	☐ No			Viability (X)	%	
Was any pump tubing shifted?	☐ Yes ☐ N/A		Osmometer ID	Osmolality (X)		
			Has Dilution criteria been met? (VCD ≥ 2.1 x 10 ⁶ cells/mL) Continue to Section 16 for Medium Dilution and N/A all subsequent sample pages			☐ Yes ☐ No
Performed By/Date:			Witness By/Date:			

Attachment 8: Additional Sampling Pages (N-2)

Perform daily sampling of the bioreactor.			BGA ID:	Measurements	Result	
Day	Seed Day _____			pH (T) (X.XX)		
Time				pCO ₂ (T) (X.X)	mmHg	
Select Primary DO Probe	☐ AIT-01 / ☐ AIT-03			pO ₂ (T) (X.X)	mmHg	
Bioreactor Temp. (°C) (XX.X) (35.0 - 38.0°C)				Bioreactor Glu (X)	mg/dL	
Pressure (psig) (X.XX) (0 - 0.6 psig)				Lactate Dilution	Calculated Lactate (mmol/L) (X.X)	
pO ₂ (%) (X.X) (10.0 - 110.0%)			Lac (mmol/L) (X.X)			
Bioreactor pH (X.XX) (6.85 - 7.50)				☐ N/A	☐ N/A	
Agitation (rpm) (X) (60 - 150 rpm)			Adjustment Perform a one-point calibration if pH difference is < -0.05 or > 0.05. Take a 2nd pH sample if pH difference is < -0.10 or > 0.10. Record information from both probes. Only primary probe results will be used for processing decisions. The secondary probe results are for information only (FIO).			
Overlay Air (slpm) (X.X) (1.0 - 10.0 slpm)			Select Primary pH Probe	Bioreactor pH (X.XX)	Offline pH (T) (X.XX)	pH Difference
Air Sparge (slpm) (X.X) (0.2 - 0.8 slpm)			☐ AIT-02		-	=
			☐ AIT-04			
O ₂ Sparge (slpm) (X.X)			Is one-point calibration required based on 1 st pH sample results?			☐ Yes ☐ No
CO ₂ Sparge (slpm) (X.X)			Is a 2 nd pH sample required? If yes, go to Attachment 6: Additional pH Monitoring – pH samples' information.			☐ Yes ☐ No
Bioreactor Weight (kg) (X)			Pipettor ID (if applicable)	Measurements	Results	
Foam Check: Is Antifoam addition Needed? Proceed to Attachment 4 for manual addition.				Vi-Cell Dilution (X)		
			Vi-Cell ID:	VCD (X.X)	x 10 ⁶ cells/mL	
☐ Yes	☐ No			Viability (X)	%	
Was any pump tubing shifted?	☐ Yes ☐ N/A		Osmometer ID	Osmolality (X)		
			Has Dilution criteria been met? (VCD ≥ 2.1 x 10 ⁶ cells/mL) Continue to Section 16 for Medium Dilution and N/A all subsequent sample pages			☐ Yes ☐ No
Performed By/Date:			Witness By/Date:			

Attachment 9: Additional Sampling Pages (N-1)

Perform daily sampling of the bioreactor.			BGA ID:	Measurements	Result	
Day	Seed Day _____			pH (T) (X.XX)		
Time				pCO ₂ (T) (X.X)	mmHg	
Choose the primary DO probe	☐ AIT-01 / ☐ AIT-03			pO ₂ (T) (X.X)	mmHg	
Bioreactor Temp. (°C) (XX.X) (35.0 - 38.0°C)				Bioreactor Glu (X)	mg/dL	
Pressure (psig) (X.XX) (0 - 0.6 psig)			Lac (mmol/L) (X.X)	Lactate Dilution	Calculated Lactate (mmol/L) (X.X)	
pO ₂ (%) (X.X) (10 - 110%)				☐ N/A	☐ N/A	
Bioreactor pH (X.XX) (6.70 - 7.50)			Adjustment Perform a one-point calibration if pH difference is < -0.05 or > 0.05. Take a 2nd pH sample if pH difference is < -0.10 or > 0.10. Record information from both probes. Only primary probe results will be used for processing decisions. The secondary probe results are for information only (FIO).			
Agitation (rpm) (X) (60 - 150 rpm)			Select Primary Probe	Bioreactor pH (X.XX)	Offline pH (T) (X.XX)	pH Difference
Overlay Air (slpm) (X.X) (1 - 10 slpm)			AIT-02	-	=	
			AIT-04			
Air Sparge (slpm) (X.X) (0.5 - 5.0 slpm)			Is one-point calibration required based on 1 st pH sample results?		☐ Yes ☐ No	
O ₂ Sparge (slpm) (X.X)			Is a 2 nd pH sample required? If yes, go to Attachment 6: Additional pH Monitoring – pH samples' information.		☐ Yes ☐ No	
CO ₂ Sparge (slpm) (X.X)			Pipettor ID (if applicable)	Measurements	Results	
Bioreactor Weight (kg) (X)				Vi-Cell Dilution (X)		
			Vi-Cell ID:	VCD (X.X)	x 10 ⁶ cells/mL	
Foam Check: Is Antifoam addition Needed? Proceed to Attachment 4 for manual additions.		☐ No ☐ Yes		Viability (X)	%	
P1	P2	P3	Osmometer ID	Osmolality (X)		
psi	psi	psi				
TMP (≤ 1.0 psi)	psi		Is Perfusion Rate Change Required? Continue to Section 20.		☐ Yes ☐ No	
Was any pump tubing shifted?	☐ Yes ☐ N/A		Has Transfer Criteria been met? (VCD ≥ 26.0 x 10 ⁶ cells/mL) Continue to Section 21.		☐ Yes ☐ No	
Performed By/Date:			Witness By/Date:			

Attachment 9: Additional Sampling Pages (N-1)

Perform daily sampling of the bioreactor.			BGA ID:	Measurements	Result	
Day	Seed Day _____			pH (T) (X.XX)		
Time				pCO ₂ (T) (X.X)	mmHg	
Choose the primary DO probe	☐ AIT-01 / ☐ AIT-03			pO ₂ (T) (X.X)	mmHg	
Bioreactor Temp. (°C) (XX.X) (35.0 - 38.0°C)				Bioreactor Glu (X)	mg/dL	
Pressure (psig) (X.XX) (0 – 0.6 psig)			Lac (mmol/L) (X.X)	Lactate Dilution	Calculated Lactate (mmol/L) (X.X)	
pO ₂ (%) (X.X) (10 - 110%)				☐ N/A	☐ N/A	
Bioreactor pH (X.XX) (6.70 - 7.50)			Adjustment <i>Perform a one-point calibration if pH difference is < -0.05 or > 0.05. Take a 2nd pH sample if pH difference is < -0.10 or > 0.10. Record information from both probes. Only primary probe results will be used for processing decisions. The secondary probe results are for information only (FIO).</i>			
Agitation (rpm) (X) (60 - 150 rpm)			Select Primary Probe	Bioreactor pH (X.XX)	Offline pH (T) (X.XX)	pH Difference
Overlay Air (slpm) (X.X) (1 - 10 slpm)			AIT-02		-	=
			AIT-04			
Air Sparge (slpm) (X.X) (0.5 - 5.0 slpm)			Is one-point calibration required based on 1 st pH sample results?			☐ Yes ☐ No
O ₂ Sparge (slpm) (X.X)			Is a 2 nd pH sample required? If yes, go to Attachment 6: Additional pH Monitoring – pH samples' information.			☐ Yes ☐ No
CO ₂ Sparge (slpm) (X.X)			Pipettor ID (if applicable)	Measurements	Results	
Bioreactor Weight (kg) (X)				Vi-Cell Dilution (X)		
			Vi-Cell ID:	VCD (X.X)	x 10 ⁶ cells/mL	
Foam Check: Is Antifoam addition Needed? Proceed to Attachment 4 for manual additions.		☐ No ☐ Yes		Viability (X)	%	
P1	P2	P3	Osmometer ID	Osmolality (X)		
psi	psi	psi				
TMP (≤ 1.0 psi)		psi		Is Perfusion Rate Change Required? Continue to Section 20.		☐ Yes ☐ No
Was any pump tubing shifted?		☐ Yes ☐ N/A		Has Transfer Criteria been met? (VCD ≥ 26.0 x 10 ⁶ cells/mL) Continue to Section 21.		☐ Yes ☐ No
Performed By/Date:				Witness By/Date:		

Attachment 9: Additional Sampling Pages (N-1)

Perform daily sampling of the bioreactor.			BGA ID:	Measurements	Result	
Day	Seed Day _____			pH (T) (X.XX)		
Time				pCO ₂ (T) (X.X)	mmHg	
Choose the primary DO probe	☐ AIT-01 / ☐ AIT-03			pO ₂ (T) (X.X)	mmHg	
Bioreactor Temp. (°C) (XX.X) (35.0 - 38.0°C)				Bioreactor Glu (X)	mg/dL	
Pressure (psig) (X.XX) (0 – 0.6 psig)			Lac (mmol/L) (X.X)	Lactate Dilution	Calculated Lactate (mmol/L) (X.X)	
pO ₂ (%) (X.X) (10 - 110%)				☐ N/A	☐ N/A	
Bioreactor pH (X.XX) (6.70 - 7.50)			Adjustment <i>Perform a one-point calibration if pH difference is < -0.05 or > 0.05. Take a 2nd pH sample if pH difference is < -0.10 or > 0.10. Record information from both probes. Only primary probe results will be used for processing decisions. The secondary probe results are for information only (FIO).</i>			
Agitation (rpm) (X) (60 - 150 rpm)			Select Primary Probe	Bioreactor pH (X.XX)	Offline pH (T) (X.XX)	pH Difference
Overlay Air (slpm) (X.X) (1 - 10 slpm)			AIT-02		-	=
			AIT-04			
Air Sparge (slpm) (X.X) (0.5 - 5.0 slpm)			Is one-point calibration required based on 1 st pH sample results?			☐ Yes ☐ No
O ₂ Sparge (slpm) (X.X)			Is a 2 nd pH sample required? If yes, go to Attachment 6: Additional pH Monitoring – pH samples' information.			☐ Yes ☐ No
CO ₂ Sparge (slpm) (X.X)			Pipettor ID (if applicable)	Measurements	Results	
Bioreactor Weight (kg) (X)				Vi-Cell Dilution (X)		
			Vi-Cell ID:	VCD (X.X)	x 10 ⁶ cells/mL	
Foam Check: Is Antifoam addition Needed? Proceed to Attachment 4 for manual additions.		☐ No ☐ Yes		Viability (X)	%	
P1	P2	P3	Osmometer ID	Osmolality (X)		
psi	psi	psi				
TMP (≤ 1.0 psi)		psi		Is Perfusion Rate Change Required? Continue to Section 20 .		☐ Yes ☐ No
Was any pump tubing shifted?		☐ Yes ☐ N/A		Has Transfer Criteria been met? (VCD ≥ 26.0 x 10 ⁶ cells/mL) Continue to Section 21 .		☐ Yes ☐ No
Performed By/Date:				Witness By/Date:		

Attachment 10: Comments Section

Step No.	Comments Provide step number(s) when applicable. Initial and date each entry.	Initials/Date

Attachment 10: Comments Section (continued)

Step No.	Comments Provide step number(s) when applicable. Initial and date each entry.	Initials/Date

Attachment 10: Comments Section (continued)

Step No.	Comments	Initials/Date
	Provide step number(s) when applicable. Initial and date each entry.	

Attachment 10: Comments Section (continued)

Step No.	Comments Provide step number(s) when applicable. Initial and date each entry.	Initials/Date

Attachment 10: Comments Section (continued)

Step No.	Comments	Initials/Date
	Provide step number(s) when applicable. Initial and date each entry.	

Attachment 10: Comments Section (continued)

Step No.	Comments	Initials/Date
	Provide step number(s) when applicable. Initial and date each entry.	

Attachment 10: Comments Section (continued)

Step No.	Comments	Initials/Date
	Provide step number(s) when applicable. Initial and date each entry.	

Attachment 10: Comments Section (continued)

Step No.	Comments	Initials/Date
	Provide step number(s) when applicable. Initial and date each entry.	

Attachment 10: Comments Section (continued)

Step No.	Comments Provide step number(s) when applicable. Initial and date each entry.	Initials/Date

Attachment 10: Comments Section (continued)

Step No.	Comments	Initials/Date
	Provide step number(s) when applicable. Initial and date each entry.	

Attachment 10: Comments Section (continued)

Step No.	Comments	Initials/Date
	Provide step number(s) when applicable. Initial and date each entry.	

Attachment 10: Comments Section (continued)

Step No.	Comments	Initials/Date
	Provide step number(s) when applicable. Initial and date each entry.	

MASTER PRODUCTION RECORD - REVISION HISTORY

The following is included for reference only and is not part of the executable record.

Version	Changes	Reasons
1.0	1. This is a new Master Production Record for the 100 L Seed Bioreactor INT-034 Process 1.	1. The information contained in this document is derived from technical report PRO-0222276.

Document Approvals

Quality Approval	Jo Pierre josha.pierre@astrazeneca.com 13-Nov-2025 20:58:47 GMT+0000
Business Approval	Thomas Halligan thomas.halligan@astrazeneca.com 17-Nov-2025 16:59:47 GMT+0000
Business Approval	Eugene Jones III eugene.jones@astrazeneca.com 17-Nov-2025 20:50:18 GMT+0000
Quality Approval	Zachary Chapman Zachary.Chapman@astrazeneca.com 20-Nov-2025 20:13:58 GMT+0000
Quality Approval	Jo Pierre josha.pierre@astrazeneca.com 20-Nov-2025 20:23:40 GMT+0000